# Enterprise Valuation

From Chapter 9 of *Valuation: The Art and Science of Corporate Investment Decisions*, 2/e.

# Enterprise Valuation

## Chapter Overview

In this chapter, we review the basics of business, or enterprise, valuation. Our focus is on the *hybrid valuation approach* that combines DCF analysis with relative valuation. We decompose the enterprise- valuation problem into two steps: The first step involves the valuation of a business's planning period cash flows spanning a 3- to 10-year period, and the second step involves the calculation of the *terminal value,* which is the value of all cash flows that follow the planning period. Pure DCF valuation models use DCF analysis to analyze the value of the planning period and terminal-value cash flows, whereas the hybrid valuation model we discuss utilizes DCF analysis to value the planning period cash flows and an EBITDA multiple to estimate the terminal value.

## 1 Introduction

In this chapter, we focus our attention on what is generally referred to as enterprise valuation, which is the valuation of a business or going concern. The approach that we recommend, which we refer to as the hybrid approach, recognizes that forecasting cash flows into the foreseeable future poses a unique challenge, because most enterprises are expected to stay in business for many years. To deal with this forecasting problem, analysts typically make explicit and detailed forecasts of firm cash flows for only a limited number of years (often referred to as the *planning period*) and estimate the value of all remaining cash flows as a *terminal value,* at the end of the planning period.

The terminal value can be estimated in one of two ways. The first method is a straightforward application of DCF analysis using the Gordon growth model. As we discussed in earlier chapters, this approach requires an estimate of both a growth rate and a discount rate. The second method applies the multiples approach we discussed in the last chapter; typically, the terminal value is determined as a multiple of the projected end-of-planning-period earnings before interest, taxes, depreciation, and amortization (EBITDA). When this latter approach is used to evaluate terminal value and DCF is used to evaluate the planning period cash flows, the model is no longer a pure DCF model but becomes a hybrid approach to enterprise valuation.

The enterprise-valuation approach described in this chapter is used in a number of applications. These include: acquisitions, which we consider in the example highlighted in this chapter; initial public offerings (where firms go public and issue equity for the first time), which we described in the previous chapter; "going private" transactions, which we will consider in the next chapter; spin-offs and carve-outs (where the division of a firm becomes a legally separate entity); and, finally, the valuation of a firm's equity for investment purposes.

In most applications, analysts use a single discount rate—the weighted average cost of capital (or WACC) of the investment—to discount both the planning period cash flows and the terminal value. This approach makes sense if the financial structure and the risk of the investment are relatively stable over time. However, analysts frequently need to estimate the enterprise value of a firm that is experiencing some sort of transition, and in these cases the firm's capital structure is often expected to change over time. Indeed, firms are often acquired using a high proportion of debt, which is then paid down over time until the firm reaches what is considered an appropriate capital structure. In these cases, the assumption of a fixed WACC is inappropriate, and we recommend the use of a variant of the discounted cash flow model known as the *adjusted present value model,* which we describe later in this chapter.

Finally, it should be noted that when firms acquire existing businesses, they typically plan on making changes in the business's operating strategy. This requires that the potential acquirer value the business given both its current strategy as well as the proposed new strategy. Valuing a publicly traded firm based on its current strategy can be viewed as a reality check—if the estimated value does not equal its actual value, why not? One could then value scenarios where the firm's operating strategy is changed following the acquisition, to determine whether the new strategy creates additional value. To help answer this question, sensitivity analysis can be deployed to determine the situations in which this additional value is indeed realized.

The chapter is organized as follows: Section 2 introduces the notion of estimating a firm's or business unit's enterprise value using the hybrid/multiples approach. Section 3 introduces the adjusted present value (APV) model, which is an alternative model that does not require that the firm's capital structure remain constant over the foreseeable future. Finally, we close our discussion of enterprise valuation with summary comments in Section 4.

## 2 Using a Two-Step Approach to Estimate Enterprise Value

We noted in the introduction that forecasting firm cash flows into the foreseeable future is a challenging task, and for that reason, analysts typically break the future into two segments: a finite number of years, known as the planning period, and all years thereafter. (See the Practitioner Insight box, "Enterprise-Valuation Methods Used on Wall Street.") Consequently, the application of the DCF model to the estimation of enterprise value can best be thought of as the sum of the two terms found in Equation 1. The first term

**PRACTITIONER INSIGHT**

### Enterprise-Valuation Methods Used on Wall Street—An Interview with Jeffrey Rabel*

In broad brush terms there are three basic valuation methodologies used throughout the investment banking industry: trading or comparable company multiples, transaction multiples, and discounted cash flows. Emphasis on a particular methodology varies depending on the particular setting or type of transaction. For equity transactions such as the pricing of an initial public offering (IPO), relative valuation based on multiples of market comparables (firms in the same or related industries, as well as firms that have been involved in recent similar transactions) is the preferred approach. The reason for the emphasis on this type of valuation is that the company will have publicly traded equity and thus investors will be able to choose whether to buy said company or any of the other companies that are "peers."

A key consideration in relative valuation analysis is the selection of a set of comparable firms. For example in the Hertz IPO, we not only looked at the relative prices of car rental firms (Avis, Budget, etc.) but also considered industrial equipment leasing companies (United Rental, RSC Equipment, etc.) because this was a growing piece of Hertz's business model. We also looked at travel-related companies, as a large part of the Hertz rental car business is driven by airline travel. Some also believed that Hertz, because of its strong brand name recognition, should be compared to valuation multiples of a set of companies with strong brand recognition, including firms such as Nike or Coke. Obviously, selection of a comparable group of firms and transactions is critical when carrying out a relative valuation, because different comparable sets trade at different multiples and this affects the valuation estimate.

In merger and acquisition (M&A) analysis involving a strategic buyer (typically another firm in the same or a related industry) or a financial buyer such as a private equity firm, the second

represents the present value of a set of cash flows spanning a finite number of years, referred to as the *planning period (PP)*.

$$\text{Enterprise Value} = \begin{array}{c}\text{Present Value}\\ \text{of the Planning}\\ \text{Period (PP) Cash Flows}\end{array} + \begin{array}{c}\text{Present Value}\\ \text{of the Terminal}\\ \text{Value}\end{array} \qquad (1)$$

The second term is the present value of the estimated terminal value ($TV_{PP}$) of the firm at the end of the planning period. As such, the terminal value represents the present value of all the cash flows that are expected to be received beyond the end of the planning period. As the Technical Insight box later on this chapter illustrates, the terminal-value estimate is generally quite important and can often represent over 50% of the value of the enterprise.

## Example—Immersion Chemical Corporation Acquires Genetic Research Corporation

To illustrate our enterprise-valuation approach, consider the valuation problem faced by the Immersion Chemical Corporation in the spring of 2010. Immersion provides products and services worldwide to the life sciences industry and operates in four businesses: Human Health, Biosciences, Animal Health/Agriculture, and Specialty and Fine type of relative valuation method—transaction multiples—is used in combination with discounted cash flows (DCF). The reason for the focus on a transaction multiple is the fact that transaction multiples represent what other buyers have been willing to pay for similar companies or companies with related lines of business. The transaction multiples are used in combination with the DCF approach, as DCF allows the buyers to value the acquisition based on their forecast of its performance under their assumptions about how the business will be run.

DCF valuation methods vary slightly from one investment bank to another; however, the typical approach to enterprise valuation is a hybrid approach consisting of forecasting cash flows to evaluate near-term projections and relative valuation to estimate a terminal value. The analysis typically involves a five-year forecast of the firm's cash flows, which are discounted using an estimate of the firm's weighted average cost of capital. Then the value of cash flows that extend beyond the end of the planning period are estimated using a terminal value that is calculated based on a multiple of a key firm performance attribute such as EBITDA. The terminal-value estimate is frequently stress tested using a constant growth rate with the Gordon growth model to assess the reasonableness of the implied growth rate reflected in the terminal-value multiple.

The final valuation approach we use [is] a variant of the DCF approach that is used where an M&A transaction involves a financial sponsor (generally a private equity firm such as Blackstone, Cerberus, or KKR). Financial buyers are primarily driven by the rate of return they earn on their investor's money, so the valuation approach they use focuses on the IRR of the transaction. The basic idea is to arrive at a set of short-term cash flow forecasts that the buyer is comfortable with, estimate a terminal value at the end of the forecast period (typically 5 years), and then determining IRRs by varying the acquisition price.

*Jeffrey Rabel is a CPA and director—Global Financial Sponsors Barclays Capital in New York. Reprinted by permission of Jeffrey Rabel.

Chemicals. The company's overall strategy is to focus on niche markets that have global opportunities, develop its strong customer relations, and enhance its new-products pipeline. In an effort to expand into biopharmaceuticals, Immersion is considering the acquisition of Genetics Research Corporation (hereafter GRC), which will be integrated into Immersion's Specialty and Fine Chemicals division.

Immersion can acquire GRC for a cash payment of $100 million, which is equal to a multiple of six times GRC's 2009 earnings before interest, taxes, depreciation, and amortization (EBITDA). Immersion will also be assuming GRC's accounts payable and accrued expenses, which total $9,825,826, so the book value of the GRC acquisition would be $109,825,826 (found in Table 1). To finance the GRC acquisition, Immersion will borrow $40 million in nonrecourse debt and will need to raise $60 million in equity.[1] Based on conversations with its investment banker, Immersion's management has learned that the $100 million acquisition cost is well within the range of five to seven times EBITDA that has recently been paid for similar firms. As the analysis below indicates, this price is also in line with Immersion's own DCF analysis of GRC, given cash flow projections based on its current operating strategy.

[1] The remaining $9,825,826 represents the value of GRC's accounts payable and accrued expenses, which will be assumed by Immersion.

**Table 1 Pre- and Post-Acquisition Balance Sheets for GRC**

| | Pre-Acquisition 2009 | Post-Acquisition 2009 |
|---|---|---|
| Current assets | $ 41,865,867 | $ 41,865,867 |
| Gross property, plant, and equipment | 88,164,876 | 88,164,876 |
| Less: Accumulated depreciation | (41,024,785) | (41,024,785) |
| Net property, plant, and equipment | $ 47,140,092 | $ 47,140,092 |
| Goodwill | — | 20,819,867 |
| Total | $ 89,005,959 | $ 109,825,826 |
| Current liabilities | $ 9,825,826 | $ 9,825,826 |
| Long-term debt | 36,839,923 | 40,000,000 |
| Total liabilities | $ 46,665,749 | $ 49,825,826 |
| Common stock (par) | 290,353 | 400,000 |
| Paid-in capital | 20,712,517 | 59,600,000 |
| Retained earnings | 21,337,340 | — |
| Common equity | $ 42,340,210 | $ 60,000,000 |
| Total | $ 89,005,959 | $ 109,825,826 |

**Legend**

| | |
|---|---|
| Acquisition Financing— | |
| Assumed payables and accrued expenses | $ 9,825,826 |
| Long-term debt (40% of enterprise value) | 40,000,000 |
| Sale of common stock | 60,000,000 |
| Total | $ 109,825,826 |
| Shares issued ($1.00 par value) | 400,000 |
| Net proceeds | $ 60,000,000 |
| Price per share | $ 150.00 |

Table 1 contains the pre- and post-acquisition balance sheets for GRC, which indicate that Immersion is paying just over $20 million more than GRC's book value for the business. Note that the $20,819,867 difference between the purchase price and GRC's pre-acquisition (book value) total assets is recorded as goodwill on the revised balance sheet.[2] Moreover, GRC's post-acquisition balance sheet contains $40 million in long-term debt (on which it pays 6.5% interest) plus $60 million in common equity (common stock at par plus paid-in capital). The remainder of the right-hand side of the balance sheet consists of current liabilities that consist entirely of payables and

[2] By including all of the purchase premium as goodwill, we are assuming that the appraised value of GRC's assets is equal to their book value.

accrued expenses, which are non-interest-bearing liabilities. Thus, after it is acquired by Immersion, GRC will have a total of \$100 million in invested capital, of which 40% is debt and 60% is equity.

## Valuing GRC Using DCF Analysis

We laid out the following three-step process for using DCF analysis, which we will now apply to the valuation of GRC:

**Step 1:** Estimate the amount and timing of the expected cash flows.
**Step 2:** Estimate a risk-appropriate discount rate.
**Step 3:** Calculate the present value of the expected cash flows, or enterprise value.

Table 2 details our analysis of the DCF evaluation of the enterprise value of GRC under the assumption that the firm's operations continue as they have in the past. We refer to this as the *status quo strategy*.

**Step 1:** *Estimate the amount and timing of the expected cash flows.* In Panels a and b of Table 2, we present the pro forma financial statements and cash flow projections for GRC that are required to complete Step 1 of a DCF analysis. The cash flow projections consist of planning period cash flows spanning the period 2010–2015 and terminal-value estimates based on cash flow projections for 2016 and beyond. In addition, Panel a contains the pro forma financial statements upon which these projections are built. The pro forma income statements assume that GRC maintains its operations as it has in the past (i.e., under a status quo strategy) and reflect an assumed rate of growth in revenues of 4% per year during the planning period.

The asset levels found in the pro forma balance sheets reflect the assets that GRC needs to support the projected revenues. The initial financing for the acquisition was presented earlier in Table 1. Any additional financing requirements are assumed to be raised by first retaining 80% of GRC's earnings and then raising any additional funds the firm requires using long-term debt. A quick review of the pro forma balance sheets found in Panel a of Table 2, however, indicates that under the status quo strategy, GRC's long-term debt actually declines from \$40 million at the end of 2009 to \$23,608,049 by 2015. This decrease reflects the fact that, given our projections, the firm's retention of future earnings is more than adequate to meet its financing needs, which allows the firm to retire its long-term debt. Finally, GRC's estimated cash flows, also found in Panel a, indicate that from 2010 through 2015 cash flows are expected to grow from \$3,538,887 to \$8,263,306.

Following the planning period cash flows, Panel b includes two analyses of the terminal value of GRC, undertaken in 2015. The first method (Method #1) uses the Gordon growth model to estimate the present value of the firm's free cash flows (FCFs) beginning in 2016 and continuing indefinitely. Specifically, we estimate the terminal value in 2015 using Equation 2 below:

$$\begin{matrix}\text{Terminal} \\ \text{Value}_{2015}\end{matrix} = \frac{\text{FCF}_{2015}\left(1 + \begin{matrix}\text{Terminal} \\ \text{Growth Rate } (g)\end{matrix}\right)}{\left(\begin{matrix}\text{Cost of Capital} \\ (k_{\text{WACC}})\end{matrix} - \begin{matrix}\text{Terminal} \\ \text{Growth Rate } (g)\end{matrix}\right)} \qquad (2)$$

**Table 2** Estimating GRC's Enterprise Value Using DCF Analysis (Status Quo Strategy)

**Panel a. (Step 1) Estimate the Amount and Timing of the Planning Period Future Cash Flows**

**Planning Period Pro Forma Financial Statements**

| | Pre-Acquisition | Post-Acquisition | Pro Forma Income Statements | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2009 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Revenues | $ 80,000,000 | $ 80,000,000 | $ 83,200,000 | $ 86,528,000 | $ 89,989,120 | $ 93,588,685 | $ 97,332,232 | $ 101,225,521 |
| Cost of goods sold | (45,733,270) | (45,733,270) | (50,932,000) | (52,629,280) | (54,394,451) | (56,230,229) | (58,139,438) | (60,125,016) |
| Gross profit | $ 34,266,730 | $ 34,266,730 | $ 32,268,000 | $ 33,898,720 | $ 35,594,669 | $ 37,358,456 | $ 39,192,794 | $ 41,100,506 |
| General and administrative expense | (17,600,000) | (17,600,000) | (17,984,000) | (18,383,360) | (18,798,694) | (19,230,642) | (19,679,868) | (20,147,063) |
| Depreciation expense | (6,500,000) | (6,500,000) | (7,856,682) | (7,856,682) | (7,856,682) | (7,856,682) | (7,856,682) | (7,856,682) |
| Net operating income | $ 10,166,730 | $ 10,166,730 | $ 6,427,318 | $ 7,658,678 | $ 8,939,292 | $ 10,271,131 | $ 11,658,244 | $ 13,096,761 |
| Interest expense | (2,523,020) | (2,523,020) | (2,600,000) | (2,549,847) | (2,453,680) | (2,307,941) | (2,108,865) | (1,852,465) |
| Earnings before taxes | $ 7,643,710 | $ 7,643,710 | $ 3,827,318 | $ 5,108,831 | $ 6,485,612 | $ 7,963,190 | $ 9,547,379 | $ 11,244,296 |
| Taxes | (1,910,928) | (1,910,928) | (956,830) | (1,277,208) | (1,621,403) | (1,990,798) | (2,386,845) | (2,811,074) |
| Net income | $ 5,732,783 | $ 5,732,783 | $ 2,870,489 | $ 3,831,623 | $ 4,864,209 | $ 5,972,393 | $ 7,160,534 | $ 8,433,222 |

| | Pre-Acquisition | Post-Acquisition | Pro Forma Balance Sheets | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2009 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Current assets | $ 41,865,867 | $ 41,865,867 | 43,540,501.76 | 45,282,121.84 | 47,093,406.71 | 48,977,142.98 | 50,936,228.70 | 52,973,677.84 |
| Gross property, plant, and equipment | 88,164,876 | 88,164,876 | 96,021,558 | 103,878,240 | 111,734,922 | 119,591,604 | 127,448,286 | 135,304,968 |
| Less: Accumulated depreciation | (41,024,785) | (41,024,785) | (48,881,467) | (56,738,149) | (64,594,831) | (72,451,513) | (80,308,194) | (88,164,876) |
| Net property, plant, and equipment | $ 47,140,092 | $ 47,140,092 | $ 47,140,092 | $ 47,140,092 | $ 47,140,092 | $ 47,140,092 | $ 47,140,092 | $ 47,140,092 |
| Goodwill | — | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 |
| Total | $ 89,005,959 | $ 109,825,826 | $ 111,500,461 | $ 113,242,081 | $ 115,053,366 | $ 116,937,102 | $ 118,896,188 | $ 120,933,637 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Current liabilities | $ 9,825,826 | $ 9,825,826 | $ 9,975,651 | $ 10,131,469 | $ 10,293,520 | $ 10,462,053 | $ 10,637,327 | $ 10,819,613 |
| Long-term debt | 36,839,923 | 40,000,000 | 39,228,419 | 37,748,922 | 35,506,789 | 32,444,078 | 28,499,462 | 23,608,049 |
| Total liabilities | $ 46,665,749 | $ 49,825,826 | $ 49,204,070 | $ 47,880,392 | $ 45,800,309 | $ 42,906,131 | $ 39,136,790 | $ 34,427,661 |
| Common stock (par) | 290,353 | 400,000 | 400,000 | 400,000 | 400,000 | 400,000 | 400,000 | 400,000 |
| Paid-in capital | 20,712,517 | 59,600,000 | 59,600,000 | 59,600,000 | 59,600,000 | 59,600,000 | 59,600,000 | 59,600,000 |
| Retained earnings | 21,337,340 | — | 2,296,391 | 5,361,689 | 9,253,057 | 14,030,971 | 19,759,398 | 26,505,976 |
| Common equity | $ 42,340,210 | $ 60,000,000 | $ 62,296,391 | $ 65,361,689 | $ 69,253,057 | $ 74,030,971 | $ 79,759,398 | $ 86,505,976 |
| Total | $ 89,005,959 | $ 109,825,826 | $ 111,500,461 | $ 113,242,081 | $ 115,053,366 | $ 116,937,102 | $ 118,896,188 | $ 120,933,637 |

**Planning Period Cash Flow Estimates**

| | Projected Firm Free Cash Flows | | | | | |
|---|---|---|---|---|---|---|
| | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Net operating income | $ 6,427,318 | 7,658,678 | $ 8,939,292 | $ 10,271,131 | $ 11,656,244 | $ 13,096,761 |
| Less: Taxes | (1,606,830) | (1,914,670) | (2,234,823) | (2,567,783) | (2,914,061) | (3,274,190) |
| NOPAT | $ 4,820,489 | $ 5,744,009 | $ 6,704,469 | $ 7,703,349 | $ 8,742,183 | $ 9,822,571 |
| Plus: Depreciation | 7,856,682 | 7,856,682 | 7,856,682 | 7,856,682 | 7,856,682 | 7,856,682 |
| Less: CAPEX | (7,856,682) | (7,856,682) | (7,856,682) | (7,856,682) | (7,856,682) | (7,856,682) |
| Less: Changes in net working capital | (1,281,602) | (1,332,866) | (1,386,180) | (1,441,628) | (1,499,293) | (1,559,264) |
| Equals FFCF | $ 3,538,887 | $ 4,411,143 | $ 5,318,289 | $ 6,261,721 | $ 7,242,890 | $ 8,263,306 |

**Panel b. (Step 1—continued) Terminal-Value Cash Flow Estimate**

***Method #1—DCF Using the Gordon Growth Model***

**Terminal-Value Multiples from the Gordon Model**

| | Growth Rates (*g*) | | | |
|---|---|---|---|---|
| **Discount Rates** | **0%** | **1%** | **2%** | **3%** |
| 8.3% | 12.05 | 13.84 | 16.19 | 19.43 |
| 8.8% | 11.36 | 12.95 | 15.00 | 17.76 |
| 9.3% | 10.75 | 12.17 | 13.97 | 16.35 |
| 9.8% | 10.20 | 11.48 | 13.08 | 15.15 |

***Method #2—Multiples Using Enterprise-Value to EBITDA Ratio***

**Terminal-Value Estimates**

| Enterprise Value / EBITDA | Terminal Value |
|---|---|
| 5.00 | $104,767,215 |
| 5.50 | 115,243,936 |
| 6.00 | 125,720,658 |
| 6.50 | 136,197,379 |
| 7.00 | 146,674,101 |

*(continued)*

**Table 2** ***continued***

**Terminal-Value Estimates (for FCFs received in 2016 and beyond)**

| | Growth Rates (*g*) | | | |
|---|---|---|---|---|
| **Discount Rates** | **0%** | **1%** | **2%** | **3%** |
| 8.3% | $ 99,557,908 | $ 114,327,938 | $ 133,786,865 | $ 160,588,784 |
| 8.8% | $ 93,901,209 | $ 106,999,224 | $ 123,949,596 | $ 146,744,924 |
| 9.3% | $ 88,852,757 | $ 100,553,487 | $ 115,459,897 | $ 135,098,501 |
| 9.8% | $ 84,319,453 | $ 94,840,221 | $ 108,058,622 | $ 125,164,788 |

**Panel c. (Step 2) Estimate a "Risk-Appropriate" Discount Rate**

- *Cost of debt*—Estimated borrowing rate is 6.5% with a marginal tax of 25%, resulting in an after-tax cost of debt of 4.875%.
- *Cost of equity*—An average industry unlevered equity beta of .89 implies a levered equity beta for GRC of 1.28, assuming a target debt ratio of 40% and a debt beta of .30. Using the capital asset pricing model with a 10-year Treasury bond yield of 5.02% and a market risk premium of 5% produces an estimate of the levered cost of equity of 11.42%.
- *Weighted average cost of capital* (WACC)—Using the target debt-to-value ratio of 40%, the WACC is approximately 8.8%.

**Panel d. (Step 3) Calculate the Present Value of Future Cash Flows**

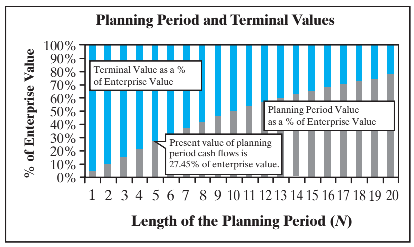
| | Present Values of Expected Future Cash Flows | | | | |
|---|---|---|---|---|---|
| | | Terminal Value | | Enterprise Value | |
| **Discount Rate** | **Planning Period FCF** | **Method #1** | **Method #2** | **Method #1** | **Method #2** |
| 7.8% | $26,201,884 | $78,986,073 | $80,112,266 | $105,187,958 | $106,314,150 |
| **8.8%** | **$25,309,857** | **$74,729,086** | **$75,794,582** | **$100,038,943** | **$101,104,439** |
| 9.8% | $24,461,738 | $70,737,378 | $71,745,960 | $ 95,199,116 | $ 96,207,698 |

## TECHNICAL INSIGHT

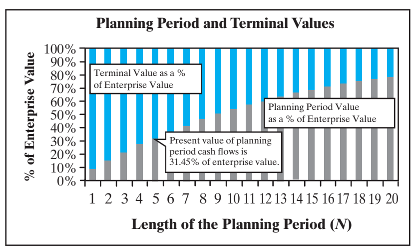
### Terminal Values, Expected Growth Rates, and the Cost of Capital

When estimating value using a planning period and a terminal value, how much of the value can be attributed to each of these components? To answer this question, consider the situation where a firm's free cash flows are expected to grow at a rate of 12% per year for a period of five years, followed by a 2% rate of growth thereafter. If the cost of capital for the firm is 10%, then the relative importance of the planning period cash flows and the terminal-value cash flows for different planning periods is captured in the upper figure.

If the analyst uses a five-year planning period, then the present value of the planning period cash flows in this setting constitutes 27.45% of the value of the enterprise, leaving over 72.55% of enterprise value in the terminal value. Similarly, if a three-year planning period is used, then the terminal value constitutes 83.83% of the enterprise-value estimate, leaving only 16.17% for the planning period. The key observation we can make from this analysis is that the terminal value is at least 50% of the value for the firm for all commonly used planning periods (i.e., 3 to 10 years).




The previous example was obviously a very high-growth firm. It stands to reason, then, that the terminal value may be less important for more stable (low-growth) firms. It turns out (as the lower figure indicates) that even for a very low- and stable-growth firm whose cash flows grow at 2% per year forever, the terminal value is still the dominant component of the enterprise-value estimate for the typical 3- to 10-year planning period. In this case, if the firm has a constant rate of growth of 2% per year for all years, a five-year planning period results in 31.45% of the enterprise value coming from the cash flows in the planning period, which leaves 68.55% of enterprise value in the terminal value.




The message is very clear. The analyst must spend significant time estimating the firm's terminal value. In fact, even for a long (by industry standards) planning period of 10 years, the terminal value for the slow-growth firm above is still roughly half (47%) of the firm's enterprise value.

How important should the terminal value be to the enterprise value of your firm? As the examples we've used above suggest, the answer will vary with the firm's growth prospects and the length of the planning period used in the analysis. In general, terminal value increases in importance with the growth rate in firm cash flows and decreases with the length of the planning period.

To estimate the terminal value using this method, we assume that the cash flows the firm is expected to generate after the end of the planning period grow at a constant rate ($g$), which is less than the cost of capital ($k_{WACC}$). Equation 2a can be interpreted as a multiple of $FFCF_{2015}$, where the multiple is equal to the ratio of 1 plus the terminal growth rate divided by the difference in the cost of capital and the growth rate, i.e.,

$$\begin{matrix}\text{Terminal} \\ \text{Value}_{2015}\end{matrix} = FCF_{2015}\left(\frac{1+g}{k_{WACC}-g}\right) = FCF_{2015} \times \left(\begin{matrix}\text{Gordon Growth} \\ \text{Model Multiple}\end{matrix}\right) \quad (2a)$$

In Panel b of Table 2, we report a panel of Gordon growth model multiples that correspond to what Immersion's management thinks is a reasonable range of values of the discount rate and rate of growth in future cash flows. For example, for an 8.8% cost of capital and a 2% terminal growth rate, the multiple for terminal value based on the Gordon growth model, $(\frac{1+g}{k_{WACC}-g})$, is 15. Based on the estimated $FCF_{2015}$ of \$8,263,306, this produces an estimate of the terminal value at the end of 2015 of \$123,949,596.

In addition to the DCF analysis of the terminal value, Panel b of Table 2 provides an analysis that uses EBITDA multiples (Method #2), as shown in Equation 3 below:

$$\begin{matrix}\text{Terminal} \\ \text{Value}_{2015}\end{matrix} = EBITDA_{2015} \times \begin{matrix}\text{EBITDA} \\ \text{Multiple}\end{matrix} \quad (3)$$

Multiples ranging from five to seven are reported in Panel b of Table 2. Using a multiple of six (which is the multiple reflected in the asking price for GRC) times $EBITDA_{2015}$ (i.e., the sum of net operating income and depreciation expense for 2015, which equals \$7,856,682) produces an estimated terminal value for GRC in 2015 of \$125,720,658 = 6 × (\$13,096,761 + 7,856,682). It should be noted that the EBITDA multiple and the free cash flow multiple generate very similar terminal-value estimates when there are no extraordinary capital expenditures or investments in net working capital. If this were not the case, the analyst would want to double-check his or her assumptions and attempt to reconcile the conflicting terminal-value estimates.

**Step 2:** *Estimate a risk-appropriate discount rate.* Because the debt financing for GRC is nonrecourse to Immersion, Immersion's finance staff analyzed the cost of capital using the project financing method described earlier. Details supporting the calculation are provided in Panel c of Table 2. The analysis resulted in an estimated cost of capital for GRC of 8.8%.

**Step 3:** *Calculate the present value of the expected cash flows, or enterprise value.* In Panel d of Table 2, we estimate the enterprise value of GRC, using the cash flow estimates from Panel a and discounting them with the estimated cost of capital for GRC from Panel c, plus and minus one percent—8.8% (the estimated WACC), 7.8%, and 9.8%. The result is an array of enterprise-value estimates reflecting each of the methods used to estimate the terminal value and the range of cost of capital estimates. With the 8.8% cost of capital, the estimate of enterprise value is just slightly higher than \$100 million. Indeed, based on the \$100 million investment, and the cash flow forecast found in Panel a of Table 2, we estimate that the internal rate of return for the acquisition is 9.02% (when the relatively conservative EBITDA multiples are used to compute the terminal value of the cash flows).

Based on this analysis, we can see that the acquisition is fairly priced, with an enterprise value very nearly equal to the $100 million in invested capital. In other words, this is essentially a zero-NPV investment. Given the uncertainties that naturally underlie the cash flow and discount rate estimates, we would not expect Immersion's managers to proceed with this acquisition unless they expect to make changes in the operating strategy of the business that will make the investment look more attractive. As we discuss below, Immersion's management does, in fact, wish to consider implementing changes.

Typically, a new owner will consider changes in operating strategies following an acquisition. In this case, Immersion's management is considering a proposal that includes additional expenditures on capital equipment and marketing, with the hope of expanding GRC's market share. Given the economies of scale associated with manufacturing in this industry, the higher market share will lead to higher operating margins and an enhanced value for the acquired business. The question we need to answer, then, is whether these anticipated changes will in fact increase the NPV of the investment.

## Valuing GRC Under the Growth Strategy

To evaluate the GRC acquisition under the assumed growth strategy, Immersion's analysts carefully revised GRC's projected cash flows and repeated the valuation analysis carried out for the status quo case. The results of this analysis are contained in Table 3. Panel a of Table 3 presents GRC's most recent year's financial statements (for 2009, both pre- and post-acquisition) and pro forma financial statements for the six-year planning period that spans 2010 through 2015.

The growth strategy involves increased expenditures on marketing as well as the addition of new manufacturing capacity. Specifically, the plan calls for spending an additional $3 million per year on advertising during 2010–2012, plus an additional $1 million per year on capital equipment during 2010–2015. Immersion's analysts expect the combined effect of these actions to double the planning period rate of growth in sales to 8% per year, compared to only 4% under the status quo strategy. After achieving the higher target market share in 2015, capital expenditures and marketing expenditures are expected to return to the status quo levels, and the expected rate of growth in free cash flows for 2016 and beyond is assumed to be 2%, just as in the status quo case.

Comparing the cash flow projections for the status quo strategy found in Panel a of Table 2 with those of the growth strategy in Panel a of Table 3, we see that the growth strategy has the initial effect of reducing cash flows below status quo levels for 2010–2012. However, beginning in 2013, the growth strategy expected cash flows exceed those of the status quo strategy ($7,331,446 for the growth strategy versus $6,261,721 for the status quo strategy) and continue to be higher for all subsequent years.

Because the growth strategy initially has lower cash flows than the status quo strategy but later generates much higher cash flows, its incremental value will depend on the discount rate that is used to evaluate the strategy. In Panel c of Table 3, we initially assume that the cost of capital for the growth strategy is the same as that for the status quo strategy (i.e., 8.8%). In this case, the enterprise-value estimates are dramatically higher than under the status quo strategy, producing an expected enterprise value of $136,273,047 when the Gordon growth model is used to estimate the terminal value (Method #1 in Panel d of Table 3) and $132,824,267 when a six-times-EBITDA multiple is used to estimate the terminal value.

**Table 3** Estimating GRC's Enterprise Value Using DCF Analysis (Growth Strategy)

**Panel a. (Step 1) Estimate the Amount and Timing of the Planning Period Future Cash Flows**

**Planning Period Pro Forma Financial Statements**

| | Pre-Acquisition | Post-Acquisition | Pro Forma Income Statements | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2009 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Revenues | $ 80,000,000 | $ 80,000,000 | $ 86,400,000 | $93,312,000 | $ 100,776,960 | $ 108,839,117 | $ 117,546,246 | $ 126,949,946 |
| Cost of goods sold | (45,733,270) | (45,733,270) | (52,564,000) | (56,089,120) | (59,896,250) | (64,007,950) | (68,448,586) | (73,244,472) |
| Gross profit | $ 34,266,730 | $ 34,266,730 | $ 33,836,000 | $37,222,880 | $ 49,880,710 | $ 44,831,167 | $ 9,097,661 | $ 53,705,473 |
| General and administrative expense | (17,600,000) | (17,600,000) | (21,368,000) | (22,197,440) | (23,093,235) | (21,060,694) | (22,105,550) | (23,233,994) |
| Depreciation expense | (6,500,000) | (6,500,000) | (7,856,682) | (8,023,349) | (8,190,015) | (8,356,682) | (8,523,349) | (8,690,015) |
| Net operating income | $ 10,166,730 | $ 10,166,730 | $ 4,611,318 | $ 7,002,091 | $ 9,597,460 | $ 15,413,791 | $ 18,468,762 | $ 21,781,465 |
| Interest expense | (2,523,020) | (2,523,020) | (2,600,000) | (2,778,968) | (2,887,535) | (2,916,242) | (2,737,729) | (2,453,086) |
| Earnings before taxes | $ 7,643,710 | $ 7,643,710 | $ 2,011,318 | $ 4,223,123 | $ 6,709,925 | $ 12,497,549 | $ 15,731,034 | $ 19,328,378 |
| Taxes | (1,910,928) | (1,910,928) | (502,830) | (1,055,781) | (1,677,481) | (3,124,387) | (3,932,758) | (4,832,095) |
| Net income | $ 5,732,783 | $ 5,732,783 | $ 1,508,489 | $ 3,167,342 | $ 5,032,444 | $ 9,373,162 | $ 11,798,275 | $ 14,496,284 |

| | Pre-Acquisition | Post-Acquisition | Pro Forma Balance Sheets | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2009 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Current assets | $ 41,865,867 | $ 41,865,867 | 45,215,137 | 48,832,347 | 52,738,935 | 56,958,050 | 61,514,694 | 66,435,870 |
| Gross property, plant, and equipment | 88,164,876 | 88,164,876 | 97,021,558 | 106,044,907 | 115,234,922 | 124,591,604 | 134,114,953 | 142,804,968 |
| Less: Accumulated depreciation | (41,024,785) | (41,024,785) | (48,881,467) | (56,904,815) | (65,094,831) | (73,451,513) | (81,974,861) | (90,664,876) |
| Net property, plant, and equipment | $ 47,140,092 | $ 47,140,092 | $ 48,140,092 | $ 49,140,092 | $ 50,140,092 | $ 51,140,092 | $ 52,140,092 | $ 52,140,092 |
| Goodwill | — | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 | 20,819,867 |
| Total | $ 89,005,959 | $ 109,825,826 | $ 114,175,095 | $ 118,792,306 | $ 123,698,894 | $ 128,918,009 | $ 134,474,653 | $ 139,395,829 |

| | Pre-Acquisition | Post-Acquisition | Pro Forma Balance Sheets | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2009 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Current liabilities | $ 9,825,826 | $ 9,825,826 | $ 10,214,944 | $ 10,628,033 | $ 11,067,013 | $ 11,533,953 | $ 12,031,091 | $ 12,471,376 |
| Long-term debt | 36,839,923 | 40,000,000 | 42,753,361 | 44,423,608 | 44,865,262 | 42,118,906 | 37,739,792 | 30,623,656 |
| Total liabilities | $ 46,665,749 | $ 49,825,826 | $ 52,968,305 | $ 55,051,642 | $ 55,932,274 | $ 53,652,860 | $ 49,770,884 | $ 43,095,032 |
| Common stock (par) | 290,353 | 340,353 | 340,353 | 340,353 | 340,353 | 340,353 | 340,353 | 340,353 |
| Paid-in capital | 20,712,517 | 38,322,307 | 38,322,307 | 38,322,307 | 38,322,307 | 38,322,307 | 38,322,307 | 38,322,307 |
| Retained earnings | 21,337,340 | 21,337,340 | 22,544,131 | 25,078,005 | 29,103,960 | 36,602,489 | 46,041,110 | 57,638,136 |
| Common equity | $ 42,340,210 | $ 60,000,000 | $ 61,206,791 | $ 63,740,665 | $ 67,766,620 | $ 75,265,149 | $ 84,703,769 | $ 96,300,796 |
| Total | $ 89,005,959 | $ 109,825,826 | $ 114,175,095 | $ 118,792,306 | $ 123,698,894 | $ 128,918,009 | $ 134,474,653 | $ 139,395,829 |

**Planning Period Cash Flow Estimates**

| | Projected Free Cash Flows | | | | | |
|---|---|---|---|---|---|---|
| | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Net operating income | $ 4,611,318 | 7,002,091 | $ 9,597,460 | $ 15,413,791 | $ 18,468,762 | $ 21,781,465 |
| Less: Taxes | (1,152,830) | (1,750,523) | (2,399,365) | (3,853,448) | (4,617,191) | (5,445,366) |
| NOPAT | $ 3,458,489 | $ 5,251,569 | $ 7,198,095 | $ 11,560,343 | $ 13,851,572 | $ 16,336,098 |
| Plus: Depreciation | 7,856,682 | 8,023,349 | 8,190,015 | 8,356,682 | 8,523,349 | 8,690,015 |
| Less: CAPEX | (8,856,682) | (9,023,349) | (9,190,015) | (9,356,682) | (9,523,349) | (8,690,015) |
| Less: Changes in net working capital | (2,563,203) | (2,768,260) | (2,989,720) | (3,228,898) | (3,487,210) | (3,766,187) |
| Equals FCF | $ (104,715) | $ 1,483,309 | $ 3,208,375 | $ 7,331,446 | $ 9,364,362 | $ 12,569,912 |

(*continued*)

**Table 3** *continued*

**Panel b. (Step 1—continued) Terminal-Value Cash Flow Estimate**

*Method #1—DCF Using the Gordon Growth Model*

**Terminal-Value Estimates (for FCFs received in 2016 and beyond)**

| Discount Rates | Growth Rates ($g$) 0% | 1% | 2% | 3% |
|---|---|---|---|---|
| 7.8% | $161,152,717 | $186,700,163 | $221,057,072 | $269,729,361 |
| 8.8% | $142,839,909 | $162,764,244 | $188,548,679 | $223,224,298 |
| 9.8% | $128,264,408 | $144,268,308 | $164,375,772 | $190,397,196 |
| 10.8% | $116,388,074 | $129,547,052 | $145,696,707 | $165,987,299 |

*Method #2—Multiples Using Enterprise Value to EBITDA Ratio*

**Terminal-Value Estimates**

| Enterprise Value / EBITDA | Terminal Value |
|---|---|
| 5.00 | $ 152,356,701 |
| 5.50 | $ 167,592,441 |
| 6.00 | $ 182,828,181 |
| 6.50 | $ 198,063,921 |
| 7.00 | $ 213,299,661 |

**Panel c. (Step 2) Estimate a "Risk-Appropriate" Discount Rate**

Based on the analysis presented in Panel c of Table 1, GRC's WACC is estimated to be 8.8%.

**Panel d. (Step 3) Calculate the Present Value of Future Cash Flows**

| | Present Values of Expected Future Cash Flows | | | | |
|---|---|---|---|---|---|
| | | Terminal Value | | Enterprise Value | |
| Discount Rate | Planning Period Firm FCF | Method #1 | Method #2 | Method #1 | Method #2 |
| 7.8% | $23,611,874 | $140,862,951 | $116,502,571 | $164,474,826 | $140,114,445 |
| 8.8% | $22,600,651 | $113,672,395 | $110,223,616 | $136,273,047 | $132,824,267 |
| 9.8% | $21,643,890 | $83,145,843 | $104,335,942 | $104,789,733 | $125,979,832 |

However, the growth strategy is almost certainly more risky than the status quo strategy and should require a higher cost of capital. To evaluate how a higher cost of capital will affect the value of the growth strategy, Panel d of Table 3 presents values of the growth strategy cash flows with alternative discount rates as high as 12%. Although the NPV is reduced, it remains positive even using a 12% discount rate, which represents an increase of over 30% in the 8.8% cost of capital we used to value the status quo strategy.

## Sensitivity Analysis

The acquisition of GRC is a risky investment, so it is important that we perform a sensitivity analysis of the proposal. In this instance, we will limit ourselves to the use of breakeven sensitivity analysis, although, it is also useful to examine a simulation model that calculates the influence of the key value drivers.

We consider three important value drivers for the GRC acquisition under the growth strategy. The first relates to the cost of capital for the acquisition, and the second and third pertain to determinants of the level of the future cash flows: (1) the estimated rate of growth in GRC's cash flows during the planning period and (2) the terminal-value multiple used to value the post-planning-period cash flows. In the risk analysis—that follows, we focus our attention solely on the valuation that uses the EBITDA multiple (Method #2) to estimate the terminal value.

### Sensitivity Analysis—Cost of Capital

The cost of capital, like all the value drivers, is always estimated with some error. However, in this instance, we have another reason to be concerned about the cost of capital estimate. The growth strategy is a riskier strategy than the status quo strategy, which means that the growth strategy cash flows should require a higher discount rate—but how much higher? One approach we can take to addressing this issue is to explore the importance of the discount rate to the valuation of GRC under the growth strategy. To do this, we can calculate the internal rate of return (IRR) of the investment and ask whether it is plausible that the appropriate discount rate exceed the IRR. Recall that we considered a similar question when evaluating the status quo strategy, where the IRR was only 9.02% and the cost of capital was 8.8%. In that instance, there was little room for error in estimating the WACC.

We calculate the IRR for the acquisition based on the expected cash flows found in Panel a in Table 3 (and a terminal value for 2015 equal to six times $EBITDA_{2015}$) and the $100 million in invested capital reflected in the asking price. Based on these assumptions, the investment has an IRR of 14.28%. Consequently, if the higher cost of capital for the riskier growth strategy exceeds 14.28%, the acquisition should not be undertaken. Although the appropriate discount rate for this investment is likely to be higher than the discount rate for the status quo investment, it is quite unlikely that it exceeds 14.28%. In Panel c of Table 2, we noted that GRC's estimated equity beta was 1.28, which generates a cost of equity of 11.42% and a weighted average cost of capital of 8.8%. To generate a cost of capital of 14.28%, GRC's equity beta would have to rise to 3.11, which is highly unlikely.

### Sensitivity Analysis—Revenue Growth Rate

Next we consider a breakeven sensitivity analysis of the planning period rate of growth in revenues. This analysis reveals that the 8% expected rate of growth in revenues can be reduced to only 5% (recall that the status quo strategy assumes a 4% rate of growth in revenues) before the acquisition value drops to the $100 million purchase price.[3] The rate of growth in revenues and, consequently, firm earnings is very important because it not only determines what the annual cash flows will be during the planning period but also has an important impact on terminal value by increasing the level of $EBITDA_{2015}$.

### Sensitivity Analysis—Terminal-Value EBITDA Multiple

The final value driver we consider is the terminal-value multiple (i.e., enterprise value divided by EBITDA) that is used to estimate the terminal value of GRC in 2015. In our earlier analysis, we used a multiple of six, which was the purchase price multiple Immersion must pay for GRC. However, if we reduce this terminal-value EBITDA multiple to 4.21, the present value of the acquisition drops to the $100 million acquisition price for GRC.

## Scenario Analysis

Up to this point, we have considered three value drivers (the discount rate, the growth rate, and the EBITDA multiple at the terminal date)—one at a time. This discussion suggests that our conclusion about the attractiveness of the investment is not likely to change if we alter any one of the value drivers individually. However, there are scenarios in which all three value drivers differ from their expected values, thus resulting in a negative NPV for the GRC investment. For example, although we argued that a 14% discount rate is very unlikely, a 10% discount rate is plausible. Similarly, one might assume that the planning period revenue growth rate may be 6% rather than 8%. If these changes are made, then the terminal-value EBITDA multiple only has to drop to 5.73 times before the enterprise value of GRC under the growth strategy drops to the $100 million acquisition price.

As the following table indicates, there are a number of plausible scenarios under which the acquisition and implementation of the growth strategy might not be value enhancing. For example, with a 5.5 EBITDA multiple for our terminal-value calculation, the acquisition is worth just over $96 million (i.e., becomes a negative-NPV investment) if we also make slightly less favorable assumptions about the discount rate (10%) and the revenue growth rate for the planning period (6%). Reviewing likely, but less

| | | Breakeven Scenarios | | | Negative-NPV Scenario |
|---|---|---|---|---|---|
| **Value Driver** | **Initial Parameters** | **Scenario #1** | **Scenario #2** | **Scenario #3** | **Scenario #4** |
| Cost of capital | 8.8% | 10% | **10.72%** | 9% | **10%** |
| Revenue growth rate during the planning period | 8% | 6% | 6% | **5.1%** | **6%** |
| Terminal-value EBITDA multiple | 6 times | **5.73 times** | 6 times | 6 times | **5.5 times** |
| Enterprise value | $132,824,688 | $100,000,000 | $100,000,000 | $100,000,000 | $96,634,832 |

[3] Note that we are holding everything constant except the revenue growth rate in this analysis. For instance, the discount rate is still assumed to be 8.8%.

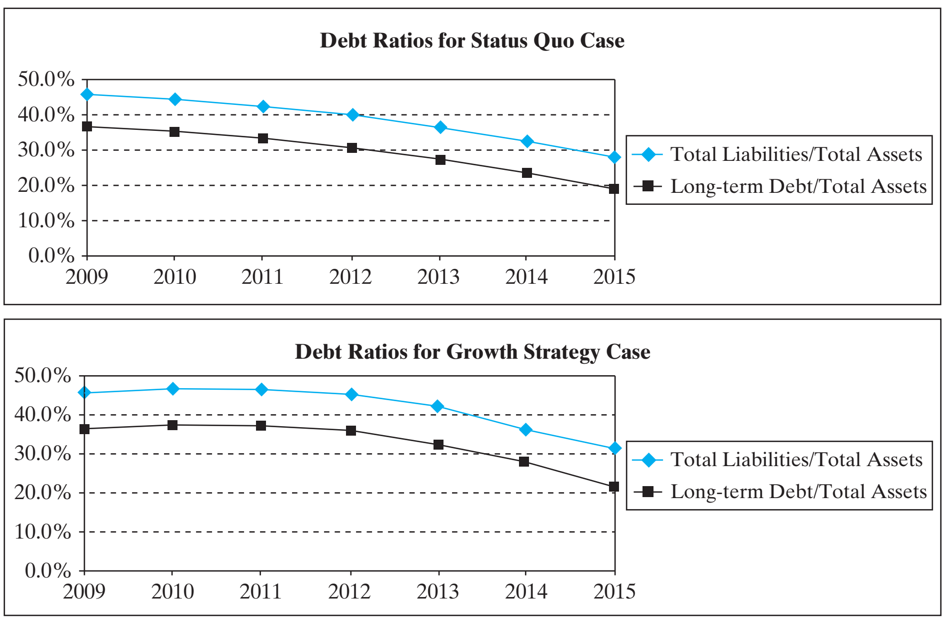
favorable, scenarios that can lead to a negative NPV is a very powerful tool for learning about an investment's risk.

The scenarios reviewed above are far from exhaustive, and we can always find scenarios under which almost any investment has either a positive or negative NPV. What this means is that the tools we have developed are just that—decision tools. They provide support and background for the actual decision maker, but they do not actually make the decision. In this particular case, the numbers do not provide a clear picture about whether or not Immersion should go forward with the acquisition. This will generally be the case. The tools provide management with valuable information, but ultimately, management must use their judgment to make the decision.

## 3 Using the APV Model to Estimate Enterprise Value

Up to this point, we have been using the **traditional WACC approach,** which uses a constant discount rate to value the enterprise cash flows. While this approach makes sense for valuing GRC prior to its acquisition by Immersion Chemical Corporation, the constant discount rate is inconsistent with the projected changes in the firm's capital structure after GRC's acquisition. A quick review of the debt ratios found in Figure 1 indicates that the

**Figure 1** **Capital Structure Ratios for the GRC Acquisition: 2009–2015**




| *Status Quo Case* | | | | | | | |
|---|---|---|---|---|---|---|---|
| | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Total Liabilities/Total Assets | 45.4% | 44.1% | 42.3% | 39.8% | 36.7% | 32.9% | 28.5% |
| Long-term Debt/Total Assets | 36.4% | 35.2% | 33.3% | 30.9% | 27.7% | 24.0% | 19.5% |
| | | | | | | | |
| *Growth Strategy Case* | | | | | | | |
| | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Total Liabilities/Total Assets | 45.4% | 46.4% | 46.3% | 45.2% | 41.6% | 37.0% | 30.9% |
| Long-term Debt/Total Assets | 36.4% | 37.4% | 37.4% | 36.3% | 32.7% | 28.1% | 22.0% |

capital structure weights (measured here in terms of book values) are not constant over time for either the status quo or growth strategy. As a result, *the use of a single discount rate is problematic.*

In situations in which the firm's capital structure is expected to substantially change over time, we recommend that the **adjusted present value,** or **APV approach,** be used.

## Introducing the APV Approach

The APV approach expresses enterprise value as the sum of the following two components:

$$\begin{array}{c}\text{Enterprise Value}\\ \text{(APV Approach)}\end{array} = \left(\begin{array}{c}\text{Value of the}\\ \text{Unlevered}\\ \text{Free Cash Flows}\end{array}\right) + \left(\begin{array}{c}\text{Value of the}\\ \text{Interest}\\ \text{Tax Savings}\end{array}\right) \tag{4}$$

The first component is the value of the firm's operating cash flows. Because the operating cash flows are not affected by how the firm is financed, we refer to these cash flows as the unlevered free cash flows. The present value of the unlevered free cash flows represents the value of the firm's cash flows under the assumption that the firm is 100% equity financed. The second component on the right-hand side of Equation 4 is the present value of the interest tax savings associated with the firm's use of debt financing. The basic premise of the APV approach is that debt financing provides a tax benefit because of the interest tax deduction.[4] By decomposing firm value in this way, the analyst is forced to deal explicitly with how the financing choice influences enterprise value.

## Using the APV Approach to Value GRC Under the Growth Strategy

The APV approach is typically implemented using a procedure very similar to what we did to estimate the enterprise value using the traditional WACC approach—found in Equation 1 and described in Equation 4a below.

$$\begin{array}{c}\text{Enterprise}\\ \text{Value}\\ \text{(APV}\\ \text{Approach)}\end{array} = \left(\begin{array}{c}\text{Value of the}\\ \text{Unlevered Free}\\ \text{Cash Flows for the}\\ \text{Planning Period}\end{array} + \begin{array}{c}\text{Value of the}\\ \text{Planning Period}\\ \text{Interest Tax}\\ \text{Savings}\end{array}\right) + \left(\begin{array}{c}\text{Present Value}\\ \text{of the Estimated}\\ \text{Terminal Value}\end{array}\right) \tag{4a}$$

[4] Technically, the second term can be an amalgam that captures all potential "side effects" of the firm's financing decisions. In addition to interest tax savings, the firm may also realize financing benefits that come in the form of below-market or subsidized financing. For example, when one firm acquires assets or an operating division from another, it is not uncommon for the seller to help finance the purchase with a very attractive loan rate.

That is, we make detailed projections of cash flows for a finite planning period and then capture the value of all cash flows after the planning period in a terminal value. From a practical perspective, the principal difference between the APV and traditional WACC approaches is that, with the APV approach, we have two cash flow streams to value: the unlevered cash flows and the interest tax savings.

Figure 2 summarizes the implementation of the APV approach in three steps: First, we estimate the value of the planning period cash flows in two components: unlevered (or operating) cash flows and interest tax savings resulting from the firm's use of debt financing. In Step 2, we estimate the residual or terminal value of the levered firm at the end of the planning period, and finally, in Step 3, we sum the present values of the planning period cash flows and the terminal value to estimate the enterprise value of the firm.

## Step 1: Estimate the Value of the Planning Period Cash Flows

The planning period cash flows are comprised of both the unlevered free cash flows and the interest tax savings. We value these cash flows for GRC using two separate calculations. As we show in Equation 5, the unlevered free cash flows are the same as the free cash flows that are discounted using the WACC approach. However, with the APV approach, these cash flows are discounted at the unlevered cost of equity rather than the WACC.

$$\begin{array}{c}\text{Value of the}\\ \text{Planning Period}\\ \text{Unlevered}\\ \text{Free Cash Flows}\end{array} = \sum_{t=1}^{PP} \frac{FCF_t}{(1 + k_{\text{Unlevered}})^t} \tag{5}$$

Applying Equation 5 to the valuation of GRC's operating cash flows for the planning period (2010–2015) for the growth strategy case summarized in Table 4, we estimate a value of \$21,955,607:

$$\begin{aligned}\begin{array}{c}\text{Value of the}\\ \text{Unlevered}\\ \text{Free Cash Flows}\end{array} &= \left( \frac{\$(104{,}715)}{(1 + .0947)^1} + \frac{\$1{,}483{,}309}{(1 + .0947)^2} + \frac{\$3{,}208{,}375}{(1 + .0947)^3} \right.\\ &\quad \left. + \frac{\$7{,}331{,}446}{(1 + .0947)^4} + \frac{\$9{,}364{,}362}{(1 + .0947)^5} + \frac{\$12{,}569{,}912}{(1 + .0947)^6} \right)\\ &= \$21{,}955{,}607\end{aligned}$$

This valuation is based on an estimate of the unlevered cost of equity equal to 9.47%. We estimate the unlevered cost of equity in Figure 3 by estimating the firm's WACC, and used again to estimate a project's cost of capital. The estimated unlevered beta for GRC is .89, which, when combined with a 10-year U.S. Treasury yield of 5.02% and a 5% market risk premium, generates an unlevered cost of equity of 9.47%.

## Figure 2 Using the Adjusted Present Value (APV) Model to Estimate Enterprise Value

**Step 1: Estimate the Value of the Planning Period Cash Flows**

***Evaluation of Operating (Unlevered) Cash Flows***

| | |
|---|---|
| *Definition:* Unlevered Free Cash Flows = Firm or Project Free Cash Flows | *Formula:* |
| Net operating income<br>Less: Taxes<br>Net operating profit after taxes (NOPAT)<br>Plus: Depreciation expense<br>Less: Capital expenditures (CAPEX)<br>Less: Increases in net working capital<br>Equals unlevered free cash flows = FCF | $\text{Value of Planning Period Unlevered Cash Flows} = \sum_{t=1}^{PP} \frac{FCF_t}{(1 + k_{Unlevered})^t}$<br>$FCF_t$ = firm free cash flows (equals equity free cash flows for the unlevered firm)<br>$k_{Unlevered}$ = discount rate for project cash flows (unlevered equity)<br>PP = planning period |

***Evaluation of the Interest Tax Savings***

| | |
|---|---|
| *Definition:* Interest Tax Savings | *Formula:* |
| Interest Tax Savings in Year $t$ = Interest Expense in Year $t \times$ Tax Rate | $\text{Value of the Interest Tax Saving} = \left[\sum_{t=1}^{PP} \frac{\text{Interest Expense}_t \times \text{Tax Rate}}{(1 + r)^t}\right]$<br>PP = planning period<br>$r$ = firm's borrowing rate |

**Step 2: Estimate the Value of the Levered Firm at the End of the Planning Period (i.e., the terminal value)**

***Evaluation of the Terminal Value***

| | |
|---|---|
| *Definition:* Terminal value of firm is equal to the enterprise value of the levered firm at the end of the planning period.<br>*Assumptions:*<br>■ After the planning period, the firm maintains a constant proportion, $L$, of debt in its capital structure.<br>■ The firm's cash flows grow at a constant rate $g$, which is less than the firm's weighted average cost of capital forever. | *Formula:* $\text{Terminal Value of the Levered Firm}_{PP} = \frac{FCF_{PP}(1 + g)}{k_{WACC} - g}$<br>$FCF_{PP}$ = firm free cash flows for the end of the planning period<br>$k_{WACC}$ = weighted average cost of capital<br>$g$ = rate of growth in FCF after the end of the planning period in perpetuity |

**Step 3: Sum the Estimated Values for the Planning Period and Terminal Period Cash Flows**

$$\text{APV Model of Enterprise Value} = \underbrace{\left[\sum_{t=1}^{PP} \frac{FCF_t}{(1 + k_{Unlevered})^t} + \sum_{t=1}^{PP} \frac{\text{Interest Expense}_t \times \text{Tax Rate}}{(1 + r)^t}\right]}_{\textbf{Step 1: Value of Planning Period Cash Flows}} + \underbrace{\left[\frac{FCF_{PP}(1 + g)}{k_{WACC} - g}\left(\frac{1}{1 + k_{Unlevered}}\right)^{PP}\right]}_{\textbf{Step 2: Value of the Terminal Period Cash Flows}}$$

**Table 4 GRC's Operating and Financial Cash Flows for the Planning Period**

| **Panel a. Unlevered Free Cash Flows (same as FCF)** | | | | | | |
|---|---|---|---|---|---|---|
| | **2010** | **2011** | **2012** | **2013** | **2014** | **2015** |
| Status quo strategy | $3,538,887 | $4,411,143 | $5,318,289 | $6,261,721 | $7,242,890 | $ 8,263,306 |
| Growth strategy | $ (104,715) | $1,483,309 | $3,208,375 | $7,331,446 | $9,364,362 | $12,569,912 |
| **Panel b. Interest Tax Savings** | | | | | | |
| | **2010** | **2011** | **2012** | **2013** | **2014** | **2015** |
| Status quo strategy | $ 650,000 | $ 637,462 | $ 613,420 | $ 576,985 | $ 527,216 | $ 463,116 |
| Growth strategy | $ 650,000 | $ 694,742 | $ 721,884 | $ 729,061 | $ 684,432 | $ 613,272 |

Next we calculate the value of the interest tax savings for the planning period as follows:

$$\begin{matrix}\text{Value of the}\\ \text{Planning Period}\\ \text{Interest Tax Savings}\end{matrix} = \sum_{t=1}^{PP} \frac{\text{Interest Expense}_t \times \text{Tax Rate}}{(1 + r)^t} \qquad (6)$$

where $r$ is the firm's borrowing rate. Substituting GRC's interest tax savings for the growth strategy (found in Table 4) into Equation 6 produces a $3,307,031 estimate for the value of its planning period interest tax savings.

$$\begin{aligned}\begin{matrix}\text{Value of the}\\ \text{Planning Period}\\ \text{Interest Tax Savings}\end{matrix} &= \left(\frac{\$650{,}000}{(1 + .065)^1} + \frac{\$694{,}742}{(1 + .065)^2} + \frac{\$721{,}884}{(1 + .065)^3}\right. \\ &\quad \left. + \frac{\$729{,}061}{(1 + .065)^4} + \frac{\$684{,}432}{(1 + .065)^5} + \frac{\$613{,}272}{(1 + .065)^6}\right) \\ &= \$3{,}307{,}031\end{aligned}$$

The combined value of operating cash flows and interest tax savings for the planning period under the growth strategy, then, is $25,262,638 = $21,955,607 + 3,307,031.

## Step 2: Estimate GRC's Terminal Value

The terminal-value calculation for the APV approach is identical to the calculation we made for our WACC analysis. As we previously stated, at the terminal date, GRC's cash flows are assumed to grow at a constant rate of 2% per year, and its capital structure will revert to a constant mix of debt and equity, keeping the debt-to-value ratio at 40%. Therefore, we can use the Gordon growth model to estimate the terminal value of the levered firm at the end of the planning period (PP) as follows:

$$\begin{matrix}\text{Terminal Value of the}\\ \text{Levered Firm}_{PP}\end{matrix} = \frac{FCF_{PP}(1 + g)}{k_{WACC} - g} \qquad (8)$$

GRC's $FFCF_{PP}$ in 2015 (which is the end of the planning period) is equal to $12,569,912 (see Table 4), and this FCF is expected to grow at a rate $g$ of 2% in perpetuity; the

**Figure 3** **Three-Step Process for Estimating GRC's Unlevered Cost of Equity**

Step 1: Identify a set of firms that operate in the same line of business as GRC. For each market comp, either estimate directly or locate published estimates of its levered equity beta, $\beta_{\text{Levered}}$, the book value of the firm's interest-bearing debt, and the market capitalization of the firm's equity.[5]

GRC: A review of the specialty chemical industry reveals four firms that are reasonably similar to GRC. Their equity betas (i.e., levered equity betas) range from .88 to 1.45, and their debt-to-equity ratios range from 35% to 80%.

| Comparable Firms | Levered Equity Betas | Debt/Equity Ratio | Assumed Debt Beta | Unlevered Equity Betas |
|---|---|---|---|---|
| A | 1.45 | 0.65 | 0.30 | 1.00 |
| B | 1.40 | 0.60 | 0.30 | 0.99 |
| C | 0.88 | 0.35 | 0.30 | 0.73 |
| D | 1.25 | 0.80 | 0.30 | 0.83 |
| Averages | 1.25 | 0.60 | 0.30 | 0.89 |

Step 2: Unlever each of the levered equity beta coefficients and average them to get an estimate of GRC's unlevered equity beta. The unlevering process uses the following relationship:[6]

$$\beta_{\text{Unlevered}} = \frac{\beta_{\text{Levered}} + \left(1 - T \times \dfrac{r_d}{1 + r_d}\right) L \beta_{\text{Debt}}}{1 + \left(1 - T \times \dfrac{r_d}{1 + r_d}\right) L} \qquad (7)$$

where $\beta_{\text{Unlevered}}$, $\beta_{\text{Levered}}$, and $\beta_{\text{Debt}}$ are the betas for the firm's unlevered and levered equity, plus its debt. The cost of debt is $r_d$, $T$ is the corporate tax rate, and $L$ is the debt-to-equity ratio.

GRC: Applying Equation 7 to the levered equity betas of the four comparable firms produces an average unlevered equity beta of .89.

[5] Typically, debt is estimated using the book value of the firm's interest-bearing liabilities (short-term notes payable, the current portion of the firm's long-term debt, plus long-term debt). Although we should technically use the market value of the firm's debt, it is customary to use book values, because most corporate debt is thinly traded if at all. The equity capitalization of the firm is estimated using the current market price of the firm's shares multiplied by the number of outstanding shares. Betas for individual stocks can be obtained from a number of published sources, including virtually all online investment information sources such as Yahoo Finance or Microsoft's MoneyCentral Web site.

[6] This relationship captures the effects of financial leverage on the firm's beta coefficient. It applies in the setting where the firm faces uncertain perpetual cash flows and corporate taxes, corporate debt is risky (i.e., debt betas are greater than zero), and the firm's use of financial leverage (i.e., the debt-to-equity ratio, $L$) is reset every period to keep leverage constant. For a derivation of Equation 7, see E. Arzac and L. Glosten, "A Reconsideration of Tax Shield Valuation," unpublished manuscript, 2004.

| **Figure 3** | ***continued*** |
|---|---|
| | Note that we have assumed a debt beta of .30 for the debt of all four comparable firms and a corporate tax rate of 25%. Research has shown that debt betas range from .20 to .40 in practice. In addition, we assume that the corporate debt yield for each of the comparable firms is equal to GRC's 6.5% borrowing rate. Where significant differences in debt ratings and, consequently, borrowing rates are present, we use firm-specific borrowing rates in the beta unlevering process described in Equation 7. |
| Step 3: | Substitute the average unlevered equity beta into the CAPM to estimate the unlevered cost of equity for the subject firm. |
| GRC: | Substituting our estimate of GRC's unlevered equity beta into the CAPM, where the risk-free rate is 5.02% and the market risk premium is 5%, we get an estimated cost of equity for an unlevered investment of 9.47%, i.e., $k_{\text{Unlevered equity}} = \text{Risk-Free Rate} + \beta_{\text{Unlevered}}\ (\text{Market Risk Premium})$ $k_{\text{Unlevered equity}} = .0502 + .89 \times .05 = .09469 \text{ or } 9.469\%$ |

weighted average cost of capital ($k_{\text{WACC}}$) is 8.8%.[7] Using Equation 8, we estimate the terminal value for GRC in 2015 as follows:

$$\text{Terminal Value of the Levered Firm}_{2015} = \frac{\$12{,}569{,}912(1 + .02)}{.088 - .02} = \$188{,}554{,}340$$

We have one remaining calculation to make in order to value GRC's terminal value. We need to discount the terminal value estimated in Equation 8 back to the present using the unlevered cost of equity, i.e.,

$$\text{Present Value of the Terminal Value}_{2009} = \left(\frac{\$12{,}569{,}912(1 + .02)}{.088 - .02}\right)\left(\frac{1}{(1 + .09469)^6}\right) = \$109{,}576{,}035$$

To complete the valuation of GRC using the APV method, we now sum the value of the planning period and terminal value in the final step.

### Step 3: Summing the Values of the Planning Period and Terminal Period

Using the APV approach, we estimate the enterprise value of the firm as the following sum:

$$\begin{array}{c}\text{Enterprise Value} \\ \text{(APV Approach)}\end{array} = \underbrace{\left[\sum_{t=1}^{PP} \frac{\text{FCF}_t}{(1 + k_{\text{Unlevered}})^t} + \sum_{t=1}^{PP} \frac{\text{Interest Expense}_t \times \text{Tax Rate}}{(1 + r)^t}\right]}_{\textbf{Value of Planning Period Cash Flows}}$$

[7] The weighted average cost of capital can also be related to the unlevered cost of equity capital by using the following relationship:

$$k_{\text{WACC}} = k_{\text{Unlevered}} - r_d(\text{Tax Rate})\frac{L}{1 + L}\left(\frac{1 + k_{\text{Unlevered}}}{1 + r_d}\right)$$

$$k_{\text{WACC}} = .0947 - .065 \times .25 \times \left(\frac{.40}{1 + .40}\right) \times \left(\frac{1 + .0947}{1 + .065}\right) = .088 \text{ or } 8.8\%$$

where $L$ is the ratio of debt to equity for the firm, which is assumed to be 40% in the GRC example.

$$+ \underbrace{\left[\frac{FCF_{PP}(1+g)}{k_{WACC}-g}\left(\frac{1}{1+k_{Unlevered}}\right)^{PP}\right]}_{\textbf{Value of the Terminal Period Cash Flows}} \tag{4a}$$

Substituting for the two components, we estimate GRC's enterprise value using the APV model to be $134,838,672, i.e.,

$$\begin{matrix}\text{Enterprise Value}\\ \text{(APV Approach)}\end{matrix} = \$21{,}955{,}607 + \$3{,}307{,}031 + \$109{,}576{,}035 = \$134{,}838{,}672$$

## Using an EBITDA Multiple to Calculate the Terminal Value

The preceding application of the APV approach to the estimation of GRC's enterprise value used the discounted cash flow (DCF) approach to value both the planning period value and the post-planning-period terminal value. What typically happens in practice, however, is that a market-based multiple is used to estimate the value of the post-planning-period cash flows. Equation 4b defines the APV approach of enterprise value as the sum of the present values of the planning period cash flows (i.e., both the unlevered cash flows of the firm[8] and the interest tax savings) plus the terminal value of the firm, which is estimated using the EBITDA multiple, i.e.,

$$\begin{matrix}\text{Enterprise Value}\\ \text{(Hybrid APV Approach)}\end{matrix} = \left(\begin{matrix}\text{Value of the Unlevered}\\ \text{Free Cash Flows}\\ \text{for the Planning Period}\end{matrix} + \begin{matrix}\text{Value of the Planning}\\ \text{Period Interest Tax}\\ \text{Savings}\end{matrix}\right) + \left(\begin{matrix}\text{Present Value of the}\\ \text{EBITDA Multiple Estimate}\\ \text{of Terminal Value}\end{matrix}\right) \tag{4b}$$

The values of the two planning period cash flow streams for GRC under the growth strategy were estimated earlier using Table 4 and equal $21,955,607 for the operating cash flows and $3,307,031 for the interest tax savings. Using the six times multiple of EBITDA from our previous analysis and the estimated EBITDA for 2015 (found using Table 4), we calculate GRC's terminal-value cash flow as follows:

$$\begin{matrix}\text{Present Value of the}\\ \text{Estimated Terminal}\\ \text{Value}_{2009}\end{matrix} = \frac{\text{Terminal Value}_{2015}}{(1+k_{Unlevered})^6} = \frac{6 \times \text{EBITDA}_{2015}}{(1+.0947)^6}$$

$$= \frac{\$182{,}828{,}880}{(1+.0947)^6} = \$106{,}248{,}753$$

EBITDA for 2015 is found using Table 3 by summing net operating income, which equals $21,781,465, and depreciation expense, which equals $8,690,015, to get $30,471,480.

[8] The unlevered cash flows of a firm are simply the cash flows that the firm would realize if it uses no debt financing. Furthermore, these unlevered cash flows are the same as the firm free cash flows (FCFs) calculated earlier to value the firm using the traditional WACC model. However, in the APV model, we discount the FCFs using the unlevered cost of equity capital (i.e., the equity discount rate appropriate for a firm that uses no debt financing) and then account for the value of the firm's interest tax savings as a separate present value calculation.

Multiplying this EBITDA estimate by 6 produces an estimate of the terminal value in 2015 of $182,828,880. Discounting the terminal value back to the present using the unlevered cost of equity, we get $106,248,753. To complete the estimate of enterprise value using the hybrid APV, we simply substitute our estimates of the values of the cash flow streams into Equation 4b as follows:

$$\begin{aligned}\text{Enterprise Value (Hybrid APV Approach)} &= \$21{,}955{,}607 + \$3{,}307{,}031 + \$106{,}248{,}753 \\ &= \$131{,}511{,}391\end{aligned}$$

This value is slightly lower than our earlier estimate because, in this case, the EBITDA multiple provided a more conservative estimate of GRC's terminal value.

Table 5 summarizes the APV estimates of enterprise value for the status quo and growth strategies, using our two methods for estimating terminal value.

**Table 5 APV Valuation Summary for GRC for Status Quo and Growth Strategies**

**Panel a. Status Quo Strategy**

***APV Estimate of Enterprise Value***

| | Present Values | | |
|---|---|---|---|
| | APV Calculation of Planning Period Cash Flows | DCF Estimates of Terminal Values | Total |
| Unlevered free cash flows | $ 24,738,517 | $ 72,033,945 | $ 96,772,462 |
| Interest tax savings | 2,830,870 | — | 2,830,870 |
| Total | $ 27,569,388 | $ 72,033,945 | **$ 99,603,332** |

***Hybrid APV Estimate of Enterprise Value***

| | Present Values | | |
|---|---|---|---|
| | APV Calculation of Planning Period Cash Flows | EBITDA Multiple Terminal Value | Total |
| Unlevered free cash flows | $ 24,738,517 | $ 73,060,734 | $ 97,799,251 |
| Interest tax savings | 2,830,870 | — | 2,830,870 |
| Total | $ 27,569,388 | $ 73,060,734 | **$ 100,630,121** |

**Panel b. Growth Strategy**

***APV Estimate of Enterprise Value***

| | Present Values | | |
|---|---|---|---|
| | APV Calculation of Planning Period Cash Flows | DCF Estimates of Terminal Values | Total |
| Unlevered free cash flows | $ 21,955,607 | $ 109,576,035 | $ 131,531,641 |
| Interest tax savings | 3,307,031 | — | 3,307,031 |
| Total | $ 25,262,638 | $ 109,576,035 | **$ 134,838,671** |

*(continued)*

**Table 5** ***continued***

| *Hybrid APV Estimate of Enterprise Value* | Present Values | | |
|---|---|---|---|
| | APV Calculation of Planning Period Cash Flows | EBITDA Multiple Terminal Value | Total |
| Unlevered free cash flows | $ 21,955,607 | $ 106,248,753 | $ 128,204,360 |
| Interest tax savings | 3,307,031 | — | 3,307,031 |
| Total | $ 25,262,638 | $ 106,248,753 | **$ 131,511,391** |

## Comparing the WACC and APV Estimates of GRC's Enterprise Value

Table 6 combines our estimates of GRC's enterprise value using both the traditional WACC approach and the APV approach. Although the estimates are not exactly the same, they are quite similar. In all cases, the GRC acquisition at a price of $100 million appears to be a good investment, and the growth strategy clearly dominates the status quo strategy.

**Table 6** **Summary of WACC and APV Estimates of GRC's Enterprise Value**

| **Status Quo Strategy** | **Traditional WACC** | **APV** |
|---|---|---|
| DCF estimate of terminal value | $ 100,038,943 | $ 99,603,333 |
| EBITDA multiple estimate of terminal value | 101,104,149 | 100,630,121 |
| **Growth Strategy** | **Traditional WACC** | **APV** |
| DCF estimate of terminal value | $ 136,276,460 | $ 134,838,672 |
| EBITDA multiple estimate of terminal value | 132,824,689 | 131,511,391 |

**Legend**

*Status quo strategy* represents a set of cash flow estimates that correspond to the current method of operations of the business.

*Growth strategy* represents cash flow estimates reflecting the implementation of an explicit plan to grow the business by making additional investments in capital equipment and changing the current method of operating the firm.

*DCF estimate of terminal value* is based on the Gordon growth model, where cash flows are expected to grow at a constant rate of 2% per year, indefinitely.

*EBITDA multiple estimate of terminal value* is based on a six times multiple of EBITDA estimated for 2015 (the end of the planning period).

In this particular application of the WACC and APV valuation approaches, the results are for all practical purposes the same. There are cases, however, where capital structure effects caused by very dramatic changes in debt financing over the life of the investment can lead to meaningful differences in the results of the two valuation approaches. In these instances, the APV approach has a clear advantage, in that it can more easily accommodate the effects of changing capital structure.

The EBITDA multiple used in the valuation of GRC produced results that were very similar to the DCF estimate, and this is purely an artifact of the particular choices made in carrying out this valuation. Due to the importance of the terminal value to the overall estimate of enterprise value, we recommend that both approaches be used and that, when selecting an EBITDA multiple, close attention be paid to recent transactions involving closely comparable firms. The advantage of the EBITDA multiple in this setting is that it ties the analysis of more distant cash flows back to a recent market transaction. However, the EBITDA multiple estimate of terminal value should be compared to a DCF estimate using the analyst's estimates of reasonable growth rates as a test of the reasonableness of the terminal-value estimate based on multiples.

## A Brief Summary of the WACC and APV Valuation Approaches

The following grid provides a summary of the salient features of the traditional WACC and APV approaches. As we have discussed, the traditional WACC method is the approach that is used in practice. However, when the capital structure of the firm being valued is likely to change over time, the APV is the preferred approach.

| | **Adjusted Present Value (APV) Method** | **Traditional WACC Method** |
|---|---|---|
| **Object of the Analysis** | *Enterprise value* as the sum of the values of:<br>■ The unlevered equity cash flows, and<br>■ Financing side effects | *Enterprise value* equal to the present value of the firm's free cash flows, discounted using the after-tax WACC |
| **Cash Flow Calculation** | ■ Unlevered free cash flows (equals firm free cash flows), plus<br>■ Interest tax savings | Firm free cash flows (FCF) |
| **Discount Rate(s)** | ■ Unlevered cash flows—cost of equity for unlevered firm, and<br>■ Interest tax savings—the yield to maturity on the firm's debt | After-tax weighted average cost of capital (WACC) |
| **How Capital Structure Effects Are Dealt with—Discount Rates, Cash Flows, or Both** | *Cash flows*—Capital structure affects the present value of the interest tax savings only. The value of the unlevered firm is not affected by the firm's use of debt financing. | *Discount rate*—The debt-equity mix affects the firm's WACC. However, the mix of debt and equity is assumed to be constant throughout the life of the investment. |

*(continued)*

***continued***

| **Technical Issues—Equations** | | |
|---|---|---|
| **Valuation Models** | *Enterprise Value*<br>$\text{Enterprise Value} = \text{Value of the Unlevered Firm} + \text{Value of Interest Tax Savings}$<br>$= \sum_{t=1}^{\infty} \frac{FCF_t^{Unlevered}}{(1 + k_{Unlevered})^t} + \sum_{t=1}^{\infty} \frac{\text{Interest} \times \text{Tax Rate}}{(1 + k_{Debt})^t}$<br>where<br>$FCF_t^{Unlevered}$ = free cash flows for an unlevered firm (equals $FCF_t$).<br>$k_u$ = cost of equity for an unlevered firm | *Enterprise Value*<br>$\text{Enterprise Value} = \sum_{t=1}^{\infty} \frac{FCF_t}{(1 + k_{WACC})^t}$<br>where<br>$FCF_t$ = firm free cash flow for $\text{year}_t$<br>$k_{WACC}$ = the firm's weighted average cost of capital |
| **User's Guide** | | |
| **Choosing the Right Model** | Although not as popular as the traditional WACC method, this approach is gaining support and is particularly attractive when valuing highly leveraged transactions, such as LBOs, where capital structure changes through time. | Most widely used method for valuing individual investment projects (capital budgeting) and entire firms. Because the discount rate (the WACC) assumes constant weights for debt and equity, this method is not well suited for situations where dramatic changes in capital structure are anticipated (e.g., LBOs). |

## Estimating the Value of Subsidized Debt Financing

In our preceding example, financing affects firm value only through its effect on interest tax savings. However, it is not uncommon for the seller of a business to offer an incentive to the buyer in the form of below-market-rate debt financing. To illustrate how we might analyze the value of below-market financing, consider the $40 million loan offer outlined in Table 7. Although the current market rate for such a loan is 6.5%, the loan carries an interest rate of 5%, which implies a 1.5% reduction in the rate of interest charged on the loan. The loan requires that the firm pay only interest, and no principal is due until maturity. Consequently, the required payments consist of $2 million per year in interest payments over the next five years, and, in the fifth year (the year in which the loan matures), the full principal amount of $40 million is due and payable.

To value the subsidy associated with the below-market-rate financing, we calculate that the present value of the payments, using the current market rate of interest, is only $37,506,592, which indicates that the financing provides the borrower with a subsidy worth $2,493,407! In essence, the seller has reduced the selling price of the firm or asset being sold by two and a half million dollars as an incentive to the buyer.

**Table 7 The Value of Subsidized Debt Financing**

| | | | | | |
|---|---|---|---|---|---|
| **Given:** | | | | | |
| Loan amount | $ 40,000,000 | | | | |
| Current market rate | 6.5% | | | | |
| Cost of debt (subsidized) | 5.0% | | | | |
| Term of loan | 5 | | | | |
| Type of loan | Interest only | | | | |
| | **2010** | **2011** | **2012** | **2013** | **2014** |
| Interest payments | $ 2,000,000 | $ 2,000,000 | $ 2,000,000 | $ 2,000,000 | $ 2,000,000 |
| Principal payment | | | | | 40,000,000 |
| Total | $ 2,000,000 | $ 2,000,000 | $ 2,000,000 | $ 2,000,000 | $ 42,000,000 |
| **Evaluation of Subsidized Loan:** | | | | | |
| Cash received | $ 40,000,000 | | | | |
| Present value of payments | 37,506,592 | | | | |
| Value of loan subsidy | $ 2,493,407 | | | | |

## 4 Summary

Using discounted cash flow analysis (DCF) to value a *firm* is a straightforward extension of its use in *project* valuation. The added complexity of firm valuation comes largely from the fact that the time horizon of firm cash flows is indefinite, while project cash flows are typically finite. In most cases, analysts solve this problem by estimating the value of cash flows in what is referred to as a *planning period,* using standard DCF analysis, and then using the method of comparables or multiples to calculate what is generally referred to as the *terminal value* of the investment. In most cases, the planning period cash flows and terminal-value estimates are discounted using the investment's weighted average cost of capital.

As we have noted, the WACC approach that we described in the first half of this chapter is widely used throughout industry to value businesses. However, it should be emphasized that this approach requires a number of implicit assumptions, which may be difficult to justify in most applications. In particular, the analysis assumes that the risks of the cash flows do not change over time and that the firm's financial structure does not change. For this reason, we would again like to add our usual caveat that the tools that we present are imperfect and should be viewed as providing support for management and not a substitute for management judgment.

The second half of the chapter considered the case where the debt ratio of the acquired business changes over time. When this is the case, the APV approach provides an improvement over the WACC approach. Although this approach is not frequently used by industry practitioners, we expect that over time it will become more popular.

## PROBLEMS

1 ENTERPRISE VALUATION Opex Capital is a small investment advisory firm located in Portland, Oregon, that has been hired by Winston Winery to estimate the value of Hilco Wines. Hilco is a small winery that is being considered for purchase by Winston. Opex has obtained the financial statements of Hilco and prepared the following estimates of the firm's free cash flow for the next five years:

| Year | Cash Flows |
|---|---|
| 1 | $100,000 |
| 2 | 120,000 |
| 3 | 135,000 |
| 4 | 150,000 |
| 5 | 175,000 |

At the end of five years, Opex has estimated that the winery should be worth approximately five times its Year 5 cash flow.

**a.** If the appropriate discount rate for valuing the winery is 15%, what is your estimate of the firm's enterprise value?
**b.** Winston Winery plans to borrow $400,000 to help finance the purchase. What is Hilco's equity worth?

2 ENTERPRISE VALUATION In the summer of 2010, Smidgeon Industries was evaluating whether or not to purchase one of its suppliers. The supplier, Carswell Manufacturing, provides Smidgeon with the raw steel Smidgeon uses to fabricate utility trailers. One of the first things that Smidgeon's management did was to forecast the cash flows of Carswell for the next five years:

| Year | Cash Flows |
|---|---|
| 1 | $1,200,000 |
| 2 | 1,260,000 |
| 3 | 1,323,000 |
| 4 | 1,389,150 |
| 5 | 1,458,608 |

Next, Smidgeon's management team looked at a group of similar firms and estimated Carswell's cost of capital to be 15%. Finally, they estimated that Carswell would be worth approximately six times its Year 5 cash flow in five years.

**a.** What is your estimate of the enterprise value of Carswell?
**b.** What is the value of the equity of Carswell if the acquisition goes through and Smidgeon borrows $2.4 million and finances the remainder using equity?

3 ENTERPRISE VALUATION—TRADITIONAL WACC MODEL Canton Corporation is a privately owned firm that engages in the production and sale of industrial chemicals primarily in North America. The firm's primary product line consists of organic solvents and intermediates for pharmaceutical, agricultural, and chemical products. Canton's management has recently been considering the possibility of taking the company public and has asked the firm's investment banker to perform some preliminary analysis of the value of the firm's equity.

To support its analysis, the investment banker has prepared pro forma financial statements for each of the next four years under the (simplifying) assumption that firm sales are flat (i.e., have a zero rate of growth), the corporate tax rate equals 30%, and capital expenditures are equal to the estimated depreciation expense.

**Canton Corporation Financials**

| | Pro Forma Balance Sheets ($000) | | | | |
|---|---|---|---|---|---|
| | **0** | **1** | **2** | **3** | **4** |
| Current assets | $ 15,000 | $ 15,000 | $ 15,000 | $ 15,000 | $ 15,000 |
| Property, plant, and equipment | 40,000 | 40,000 | 40,000 | 40,000 | 40,000 |
| Total | $ 55,000 | $ 55,000 | $ 55,000 | $ 55,000 | $ 55,000 |
| Accruals and payables | $ 5,000 | $ 5,000 | $ 5,000 | $ 5,000 | $ 5,000 |
| Long-term debt | 25,000 | 25,000 | 25,000 | 25,000 | 25,000 |
| Equity | 25,000 | 25,000 | 25,000 | 25,000 | 25,000 |
| Total | $ 55,000 | $ 55,000 | $ 55,000 | $ 55,000 | $ 55,000 |

| | Pro Forma Income Statements ($000) | | | |
|---|---|---|---|---|
| | **1** | **2** | **3** | **4** |
| Sales | $ 100,000 | $ 100,000 | $ 100,000 | $ 100,000 |
| Cost of goods sold | (40,000) | (40,000) | (40,000) | (40,000) |
| Gross profit | $ 60,000 | $ 60,000 | $ 60,000 | $ 60,000 |
| Operating expenses (excluding depreciation) | (30,000) | (30,000) | (30,000) | (30,000) |
| Depreciation expense | (8,000) | (8,000) | (8,000) | (8,000) |
| Operating income (earnings before interest and taxes) | $ 22,000 | $ 22,000 | $ 22,000 | $ 22,000 |
| Less: Interest expense | (2,000) | (2,000) | (2,000) | (2,000) |
| Earnings before taxes | $ 20,000 | $ 20,000 | $ 20,000 | $ 20,000 |
| Less: Taxes | (6,000) | (6,000) | (6,000) | (6,000) |
| Net income | $ 14,000 | $ 14,000 | $ 14,000 | $ 14,000 |

In addition to the financial information on Canton, the investment banker has assembled the following information concerning current rates of return in the capital market.

- The current market rate of interest on 10-year Treasury bonds is 7%, and the market risk premium is estimated to be 5%.

- Canton's debt currently carries a rate of 8%, and this is the rate the firm would have to pay for any future borrowing as well.
- Using publicly traded firms as proxies, the estimated equity beta for Canton is 1.60.

  **a.** What is Canton's cost of equity capital? What is the after-tax cost of debt for the firm?
  **b.** Calculate the equity free cash flows for Canton for each of the next four years. Assuming that equity free cash flows are a level perpetuity for Year 5 and beyond, estimate the value of Canton's equity. (*Hint:* Equity value is equal to the present value of the equity free cash flows discounted at the levered cost of equity.) If the market rate of interest on Canton's debt is equal to the 8% coupon, what is the current market value of the firm's debt? What is the enterprise value of Canton? (*Hint:* Enterprise value can be estimated as the sum of the estimated values of the firm's interest-bearing debt plus equity.)
  **c.** Using the market values of Canton's debt and equity calculated in part b above, calculate the firm's after-tax weighted average cost of capital. *Hint:*

$$k_{\text{WACC}} = \begin{matrix}\text{Cost of}\\ \text{Debt}\end{matrix}\left(1 - \begin{matrix}\text{Tax}\\ \text{Rate}\end{matrix}\right)\left(\frac{\text{Debt Value}}{\text{Enterprise Value}}\right) + \begin{matrix}\text{Cost of}\\ \text{Levered Equity}\end{matrix}\left(\frac{\text{Equity Value}}{\text{Enterprise Value}}\right)$$

  **d.** What are the firm free cash flows (FCFs) for Canton for Years 1 through 4?
  **e.** Estimate the enterprise value of Canton using the traditional WACC model, based on your previous answers and assuming that the FCFs after Year 4 are a level perpetuity equal to the Year 4 FCF. How does your estimate compare to your earlier estimate using the sum of the values of the firm's debt and equity?
  **f.** Based on your estimate of enterprise value, what is the value per share of equity for the firm if the firm has two million shares outstanding? (Remember that your calculations to this point have been in thousands of dollars.)

4 TRADITIONAL WACC VALUATION The owner of Big Boy Flea Market (BBFM), Lewis Redding, passed away on December 30, 2009. His 100% ownership interest in BBFM became part of his estate, on which his heirs must pay estate taxes. The IRS has hired a valuation expert, who has submitted a report stating that the business was worth approximately $20 million at the time of Mr. Redding's death. Mr. Redding's heirs believe the IRS valuation is too high and have hired you to perform a separate valuation analysis in hopes of supporting their position.

**a.** Estimate the firm free cash flows for 2010–2014, where the firm's tax rate is 25%, capital expenditures are assumed to equal depreciation expense, and there are no changes in net working capital over the period.
**b.** Value BBFM using the traditional WACC model and the following information:
  **i.** BBFM's cost of equity is estimated to be 20% and the cost of debt is 10%.
  **ii.** BBFM's management targets its long-term debt ratio at 10% of enterprise value.
  **iii.** After 2014, the long-term growth rate in the firm's free cash flow will be 5% per year.

5 ENTERPRISE VALUATION—APV MODEL This problem uses the information from Problem 3 about Canton Corporation to estimate the firm's enterprise value using the APV model.

**a.** What is the firm's unlevered cost of equity?

**b.** What are the unlevered free cash flows for Canton for Years 1 through 4? (*Hint:* The unlevered free cash flows are the same as the firm free cash flows.)
**c.** What are the interest tax savings for Canton Corp. for Years 1 through 4?

| | **Base Year** | **Forecast** | | | | |
|---|---|---|---|---|---|---|
| **Income Statement** | **2009** | **2010** | **2011** | **2012** | **2013** | **2014** |
| Sales | $ 3,417,500.00 | $ 3,700,625.00 | $ 3,983,750.00 | $ 4,266,875.00 | $ 4,550,000.00 | $ 4,833,125.00 |
| Depreciation | | 48,190.60 | 50,118.22 | 52,122.95 | 54,207.87 | 56,376.19 |
| COGS | | 1,402,798.69 | 1,496,881.13 | 1,590,963.56 | 1,685,046.00 | 1,779,128.44 |
| Other operating expenses | | 1,078,447.38 | 1,109,874.25 | 1,141,301.13 | 1,172,728.00 | 1,204,154.88 |
| EBIT | | $ 1,171,188.34 | $ 1,326,876.40 | $ 1,482,487.36 | $ 1,638,018.13 | $ 1,793,465.50 |
| Interest | | 5,110.00 | 5,110.00 | 5,110.00 | 5,110.00 | 5,860.00 |
| Earnings before taxes | | $ 1,166,078.34 | $ 1,321,766.40 | $ 1,477,377.36 | $ 1,632,908.13 | $ 1,787,605.50 |
| Income taxes | | 291,519.58 | 330,441.60 | 369,344.34 | 408,227.03 | 446,901.38 |
| Net income | | $ 874,558.75 | $ 991,324.80 | $ 1,108,033.02 | $ 1,224,681.10 | $ 1,340,704.13 |
| Preferred stock dividends | | — | — | — | — | — |
| Net income to common stock | | 874,558.75 | 991,324.80 | 1,108,033.02 | 1,224,681.10 | 1,340,704.13 |
| Common stock dividends | | 874,558.75 | 991,324.80 | 1,108,033.02 | 1,224,681.10 | 1,340,704.13 |
| Income to retained earnings | | — | — | — | — | — |

| | **Base Year** | **Forecast** | | | | |
|---|---|---|---|---|---|---|
| **Balance Sheet** | **2009** | **2010** | **2011** | **2012** | **2013** | **2014** |
| Current assets | $ 170,630.00 | $ 185,031.25 | $ 199,187.50 | $ 213,343.75 | $ 227,500.00 | $ 241,656.25 |
| Gross fixed assets | 1,204,765.00 | 1,252,955.60 | 1,303,073.82 | 1,355,196.78 | 1,409,404.65 | 1,540,780.83 |
| Accumulated depreciation | (734,750.00) | (782,940.60) | (833,058.82) | (885,181.78) | (939,389.65) | (995,765.83) |
| Net fixed assets | $ 470,015.00 | $ 470,015.00 | $ 470,015.00 | $ 470,015.00 | $ 470,015.00 | $ 545,015.00 |
| Total assets | $ 640,645.00 | $ 655,046.25 | $ 669,202.50 | $ 683,358.75 | $ 697,515.00 | $ 786,671.25 |
| Current liabilities | $ 129,645.00 | $ 144,046.25 | $ 158,202.50 | $ 172,358.75 | $ 186,515.00 | $ 200,671.25 |
| L-T debt | 51,100.00 | 51,100.00 | 51,100.00 | 51,100.00 | 51,100.00 | 58,600.00 |
| Preferred stock | — | — | — | — | — | — |
| Common stock | 1,000.00 | 1,000.00 | 1,000.00 | 1,000.00 | 1,000.00 | 68,500.00 |
| Retained earnings | 458,900.00 | 458,900.00 | 458,900.00 | 458,900.00 | 458,900.00 | 458,900.00 |
| Total common equity | $ 459,900.00 | $ 459,900.00 | $ 459,900.00 | $ 459,900.00 | $ 459,900.00 | $ 527,400.00 |
| Total liabilities & equity | $ 640,645.00 | $ 655,046.25 | $ 669,202.50 | $ 683,358.75 | $ 697,515.00 | $ 786,671.25 |

**d.** Assuming that the firm's future cash flows from operations (i.e., its FCFs) and its interest tax savings are level perpetuities for Year 5 and beyond that equal their Year 4 values, what is your estimate of the enterprise value of Canton Corp.?

**e.** Based on your estimate of enterprise value, what is the value per share of equity for the firm if the firm has two million shares outstanding? (Remember that your calculations to this point have been in thousands of dollars.)

6 APV VALUATION The following information provides the basis for performing a straightforward application of the adjusted present value (APV) model.

| **Assumptions** | | | | |
|---|---|---|---|---|
| Unlevered cost of equity | 12% | | | |
| Borrowing rate | 8% | | | |
| Tax rate | 30% | | | |
| Current debt outstanding | $200.00 | | | |
| | | **Years** | | |
| | **1** | **2** | **3** | **4 & Beyond** |
| Firm free cash flows | $100.00 | $120.00 | $180.00 | $200.00 |
| Interest-bearing debt | 200.00 | 150.00 | 100.00 | 50.00 |
| Interest expense | 16.00 | 12.00 | 8.00 | 4.00 |
| Interest tax savings | 4.80 | 3.60 | 2.40 | 1.20 |

**Legend**

*Unlevered cost of equity*—The rate of return required by stockholders in the company, where the firm has used only equity financing.

*Borrowing rate*—The rate of interest the firm pays on its debt. We also assume that this rate is equal to the current cost of borrowing for the firm or the current market rate of interest.

*Tax rate*—The corporate tax rate on earnings. We assume that this tax rate is constant for all income levels.

*Current debt outstanding*—Total interest-bearing debt, excluding payables and other forms of non-interest-bearing debt, at the time of the valuation.

*Firm free cash flows*—Net operating earnings after tax (NOPAT), plus depreciation (and other noncash charges), less new investments in net working capital less capital expenditures (CAPEX) for the period. This is the free cash flow to the unlevered firm, because no interest or principal is considered in its calculation.

*Interest-bearing debt*—Outstanding debt at the beginning of the period, which carries an explicit interest cost.

*Interest expense*—Interest-bearing debt for the period times the contractual rate of interest that the firm must pay.

*Interest tax savings*—Interest expense for the period times the corporate tax rate.

**a.** What is the value of the unlevered firm, assuming that its free cash flows for Year 5 and beyond are equal to the Year 4 free cash flow?

**b.** What is the value of the firm's interest tax savings, assuming that they remain constant for Year 4 and beyond?

**c.** What is the value of the levered firm?

**d.** What is the value of the levered firm's equity (assuming that the firm's debt is equal to its book value)?

7 APV VALUATION Clarion Manufacturing Company is a publicly held company that is engaged in the manufacture of home and office furniture. The firm's most recent income statement and balance sheet are found below ($000).

| **Income Statement** | | | |
|---|---|---|---|
| Sales | $ 16,000 | | |
| Cost of goods sold | (9,000) | | |
| Gross profit | 7,000 | | |
| Selling and administrative expenses | (2,000) | | |
| Operating income | 5,000 | | |
| Interest expense | (720) | | |
| Earnings before taxes | 4,280 | | |
| Taxes | (1,284) | | |
| Net income | $ 2,996 | | |
| **Balance Sheet** | | | |
| Current assets | $ 5,000 | Liabilities[9] | $ 7,000 |
| Fixed assets | 9,000 | Owners' equity | 7,000 |
| | $ 14,000 | | $ 14,000 |

**a.** If Clarion's future operating earnings are flat (i.e., no growth) and it anticipates making capital expenditures equal to depreciation expense with no increase anticipated in the firm's net working capital, what is the value of the firm using the adjusted present value model? To respond to this question, you may assume the following: The firm's current borrowing rate is the same as the 9% rate it presently pays on its debt; all the firm's liabilities are interest bearing; the firm's "unlevered" cost of equity is 12%; and the firm's tax rate is 30%.

**b.** What is the value of Clarion's equity (i.e., its levered equity) under the circumstances described above?

**c.** What is Clarion's weighted average cost of capital, given your answers to parts a and b above?

**d.** Based upon your answers to the above questions, what is Clarion's levered cost of equity?[10]

**e.** If the risk-free rate is 5.25% and the market risk premium is 7%, what is Clarion's levered beta? What is Clarion's unlevered beta?

8 TERMINAL-VALUE ANALYSIS Terminal value refers to the valuation attached to the end of the planning period and that captures the value of all subsequent cash flows. Estimate the value today for each of the following sets of future cash flow forecasts:

**a.** Claymore Mining Company anticipates that it will earn firm free cash flows (FCFs) of $4 million per year for each of the next five years. Moreover, beginning in Year 6, the firm will earn FCF of $5 million per year for the indefinite future. If Claymore's cost of capital is 10%, what is the value of the firm's future cash flows?

[9] All liabilities are assumed to be interest bearing.

[10] A firm's "levered cost of equity" represents the rate of return investors require when investing in the firm's common equity, given the firm's present use of financial leverage.

**b.** Shameless Commerce Inc. has no outstanding debt and is being evaluated as a possible acquisition. Shameless's FCFs for the next five years are projected to be $1 million per year, and, beginning in Year 6, the cash flows are expected to begin growing at the anticipated rate of inflation, which is currently 3% per annum. If the cost of capital for Shameless is 10%, what is your estimate of the present value of the FCFs?

**c.** Dustin Electric Inc. is about to be acquired by the firm's management from the firm's founder for $15 million in cash. The purchase price will be financed with $10 million in notes that are to be repaid in $2 million increments over the next five years. At the end of this five-year period, the firm will have no remaining debt. The FCFs are expected to be $3 million a year for the next five years. Beginning in Year 6, the FCFs are expected to grow at a rate of 2% per year into the indefinite future. If the unlevered cost of equity for Dustin is approximately 15% and the firm's borrowing rate on the buyout debt is 10% (before taxes at a rate of 30%), what is your estimate of the value of the firm?

9 ENTERPRISE VALUATION—TRADITIONAL WACC VERSUS APV Answer the following questions for the Canton Corporation, described in Problem 3, where firm revenues grow at a rate of 10% per year during Years 1 through 4 before leveling out at no growth for Year 5 and beyond. You may assume that *Canton's gross profit margin is 60%, operating expenses (before depreciation) to sales is 30%, current assests to sales is 15%, accrued payable to sales is 5%, PP&E to sales is 40%, and that Canton maintains* equal dollar amounts of long-term debt and equity to finance its growing needs for invested capital. Also assume that the cost of unlevered equity in this case is 13.84% and the cost of levered equity is 15.28%. The cost of debt remains at 8%. The corporate tax rate is 30%.

**a.** Calculate the enterprise value using the adjusted present value method.

**b.** From part a, calculate the value of equity after deducting the value of book debt from enterprise value. Use the market value of equity and book value of debt as weights to compute the weighted average cost of capital.

**c.** Value the firm's free cash flows using the WACC approach.

**d.** Compare your enterprise-value estimates for the two discounted cash flow models. Which of the two models do you feel best suits the valuation problem posed for Canton?

10 TERMINAL VALUE AND THE LENGTH OF THE PLANNING PERIOD The Prestonwood Development Corporation has projected its cash flows (i.e., firm free cash flows) for the indefinite future under the following assumptions: (1) Last year the firm's FCF was $1 million, and the firm expects this to grow at a rate of 20% for the next 8 years; (2) beginning Year 9, the firm anticipates that its FCF will grow at a rate of 4% indefinitely; and (3) the firm estimates its cost of capital to be 12%. Based on these assumptions, the firm's projected FCF over the next 20 years is the following:

| Years | Cash Flows (millions) |
|---|---|
| 1 | $ 1.2000 |
| 2 | 1.4400 |
| 3 | 1.7280 |
| 4 | 2.0736 |
| 5 | 2.4883 |

| | |
|---|---|
| 6 | 2.9860 |
| 7 | 3.5832 |
| 8 | 4.2998 |
| 9 | 4.4718 |
| 10 | 4.6507 |
| 11 | 4.8367 |
| 12 | 5.0302 |
| 13 | 5.2314 |
| 14 | 5.4406 |
| 15 | 5.6583 |
| 16 | 5.8846 |
| 17 | 6.1200 |
| 18 | 6.3648 |
| 19 | 6.6194 |
| 20 | 6.8841 |

**a.** Based on the information given and a terminal-value estimate for the firm of $89.4939 million for Year 20, what is your estimate of the firm's enterprise value today (in 2010)?

**b.** If a three-year planning period is used to value Prestonwood, what is the value of the planning period cash flows and the present value of the terminal value at the end of Year 3?

**c.** Answer part b for planning periods of 10 and 20 years. How does the relative importance of the terminal value change as you lengthen the planning period?

## PROBLEM 11 **MINI-CASE** APV VALUATION[11]

Flowmaster Forge Inc. is a designer and manufacturer of industrial air-handling equipment that is a wholly owned subsidiary of Howden Industrial Inc. Howden is interested in selling Flowmaster to an investment group formed by company CFO Gary Burton.

Burton prepared a set of financial projections for Flowmaster under the new ownership. For the first year of operations, firm revenues were estimated to be $160 million, variable and fixed operating expenses (excluding depreciation expense) were projected to be $80 million, and depreciation expense was estimated to be $15 million. Revenues and expenses were projected to grow at a rate of 4% per year in perpetuity.

Flowmaster currently has $125 million in debt outstanding that carries an interest rate of 6%. The debt trades at par (i.e., at a price equal to its face value). The investment group intends to keep the debt outstanding after the acquisition is completed, and the level of debt is expected to grow by the same 4% rate as firm revenues.

[11] © 2010 Scott Gibson, Associate Professor of Finance, College of William and Mary, Williamsburg, Virginia.

Projected income statements for the first three years of operation of Flowmaster following the acquisition are as follows ($ millions):

| | Pro Forma Income Statements | | |
|---|---|---|---|
| | **Year 1** | **Year 2** | **Year 3** |
| Revenues | $ 160.00 | $ 166.40 | $ 173.06 |
| Expenses | (80.00) | $ (83.20) | $ (86.53) |
| Depreciation (note 1) | (15.00) | $ (15.60) | $ (16.22) |
| Earnings before interest and taxes | $ 65.00 | $ 67.60 | $ 70.30 |
| Interest expense (note 2) | (7.50) | (7.80) | (8.11) |
| Earnings before taxes | $ 57.50 | $ 59.80 | $ 62.19 |
| Taxes (34%) | (19.55) | (20.33) | (21.15) |
| Net income | $ 37.95 | $ 39.47 | $ 41.05 |

*Note 1*—Property, plant, and equipment grow at the same rate as revenues, such that depreciation expenses grow at 4% per year.

*Note 2*—The initial debt level of $125 million is assumed to grow with firm assets at a rate of 4% per year.

Burton anticipates that efficiency gains can be implemented that will allow Flowmaster to reduce its needs for net working capital. Currently, Flowmaster has net working capital equal to 30% of anticipated revenues for Year 1. He estimates that for Year 1, the firm's net working capital can be reduced to 25% of Year 2 revenues, then 20% of revenues for all subsequent years. Estimated net working capital for Years 1 through 3 is as follows ($ millions):

| | **Current** | **Pro Forma** | | |
|---|---|---|---|---|
| Net working capital ($t-1$)/Revenues ($t$) | 30% | 25% | 20% | 20% |
| Net working capital | $ 48.00 | $ 41.60 | $ 34.61 | $ 36.00 |

To sustain the firm's expected revenue growth, Burton estimates that annual capital expenditures that equal the firm's annual depreciation expense will be required.

Gary Burton had been thinking for some time about whether to use Howden's corporate cost of capital of 9% to value Flowmaster and had come to the conclusion that an independent estimate should be made. To make the estimate, he collected the following information on the betas and leverage ratios for three publicly traded firms with manufacturing operations that are very similar to Flowmaster's:

| **Company** | **Leveraged Equity Beta** | **Debt Beta** | **Leverage Ratio*** | **Revenues** ($ millions)** |
|---|---|---|---|---|
| Gopher Forge | 1.61 | 0.52 | 0.46 | $ 400 |
| Alpha | 1.53 | 0.49 | 0.44 | 380 |
| Global Diversified | 0.73 | 0.03 | 0.15 | 9,400 |

*The leverage ratio is the ratio of the market value of debt to the market value of debt and equity.

**Revenues are the entire firm's revenues for the most recent fiscal year.

**a.** Calculate the unlevered cash flows (i.e., the firm free cash flows for Flowmaster for Years 1–3).
**b.** Calculate the unlevered cost of equity capital for Flowmaster. The risk-free rate of interest is 4.5% and the market risk premium is estimated to be 6%.
**c.** Calculate the value of Flowmaster's unlevered business.
**d.** What is the value of Flowmaster's interest tax savings, based on the assumption that the $125 million in debt remains outstanding (i.e., the investment group assumes the debt obligation) and that the firm's debt and consequently its interest expenses grow at the same rate as revenues?
**e.** What is your estimate of the enterprise value of Flowmaster based on your analysis to this point? How much is the equity of the firm worth today, assuming the $125 million in debt remains outstanding?
**f.** In conversations with the investment banker who was helping the investment group finance the purchase, Mr. Burton learned that Flowmaster has sufficient debt capacity to issue additional debt that would have a subordinate claim to the present debt holders, carrying an 8.5% rate. The amount of new debt is constrained by the need to maintain an interest coverage ratio (i.e., earnings before interest, taxes, and depreciation divided by interest expense) of five to one. Assuming that Flowmaster's $125 million of 6% senior debt remains in place (and grows at a rate of 4% per year going forward), what is the maximum amount of subordinated debt that can be issued to help finance the purchase of Flowmaster?
**g.** If the investment group decides to raise the additional subordinated debt (from part f), Gary Burton, CFO, anticipates that Flowmaster's unsecured suppliers will grant less favorable credit terms (note that the amount of subordinated debt that Flowmaster can support will continue to increase along with Flowmaster's EBITDA growth). In fact, he estimates that the decrease in accounts payable will result in an increase in the required net working capital in perpetuity by 20% of anticipated revenues per year. Consequently, the current net working capital balance would increase from 30% of revenues to 50%, or from $.3 \times \$160$ million = $48 million to $.50 \times \$160$ million = $80 million. Similarly, the projected net working capital balance for the end of Year 1 would increase from 25% to 45% of revenues. For Year 2 and beyond, working capital would increase from the 20% projected earlier to 40% of revenues. Use the APV approach to estimate Flowmaster's enterprise value to determine whether the additional subordinated debt should be issued.

## PROBLEM 12 **MINI-CASE** TRADITIONAL WACC VALUATION[12]

**Setting:** It is early January 2010, and as the chief financial officer of TM Toys Inc., you are evaluating a strategic acquisition of Toy Co. Inc. (the "target").

**Industry Overview:** The toys-and-games industry consists of a select group of global players. The $60 billion industry (excluding videos) is dominated by two U.S. toy makers: Mattel (Barbie, Hot Wheels, Fisher-Price) and Hasbro (G.I. Joe, Tonka, Playskool). International players include Japan's Bandai Co. (Digimon) and Sanrio (Hello Kitty), as well as Denmark's LEGO Holding. Success in this industry is dependent on creating

[12] 

**Exhibit P12.1** **Planning Period Cash Flow Estimates**

| Toy Co. Inc. ($ in Millions) | Projected Firm Free Cash Flows | | | | |
|---|---|---|---|---|---|
| **Fiscal Year Ended** | **12/31/10** | **12/31/11** | **12/31/12** | **12/31/13** | **12/31/14** |
| Net Operating Income | $733.16 | $757.63 | $783.64 | $799.32 | $ 815.30 |
| Less: Taxes | 201.27 | 207.98 | 235.09 | 239.80 | 244.59 |
| NOPAT | $531.90 | $549.65 | $548.55 | $559.52 | $ 570.71 |
| Plus: Depreciation | 183.58 | 186.21 | 191.80 | 195.64 | 199.55 |
| Less: Capital Expenditures | (180.00) | (212.82) | (219.20) | (223.59) | (228.06) |
| (Increase) in Working Capital | (50.37) | 43.54 | (27.68) | (19.82) | (20.21) |
| **Equals FCFF** | **$485.11** | **$566.59** | **$493.47** | **$511.76** | **$ 521.99** |
| **EBITDA** | **$916.74** | **$943.84** | **$975.45** | **$994.95** | **$1,014.85** |

cross-culturally appealing brands backed by successful marketing strategies. Toy companies achieve success through scoring the next big hit with their target consumers and unveiling the "must-have" toys. Historically, we have seen significant merger and acquisition activity and consolidation among brands in this industry.

**Target Company Description:** Toy Co. Inc. is a multibrand company that designs and markets a broad range of toys and consumer products. The product categories include: Action Figures, Art Activity Kits, Stationery, Writing Instruments, Performance Kites, Water Toys, Sports Activity Toys, Vehicles, Infant/Pre-School, Plush, Construction Toys, Electronics, Dolls, Dress-Up, Role Play, and Pet Toys and Accessories. The products are sold under various brand names. The target designs, manufactures, and markets a variety of toy products worldwide through sales to retailers and wholesalers and directly to consumers. Its stock price closed on 12/31/09 at $19.49 per share.

**Exhibit P12.2** **Estimate a "Risk-Appropriate" Discount Rate**

- Cost of debt—Estimated borrowing rate is 6.125% with a marginal tax of 27.29%, resulting in an after-tax cost of debt of 4.5%.
- Cost of equity—Levered equity beta for Toy Co. is .777; using the capital asset pricing model with a 10-year Treasury bond yield of 4.66% and a market risk premium of 7.67% produces an estimate of the levered cost of equity of 10.57%.
- Other—Diluted Shares of Common Equity outstanding on 12/31/09: 422,040,500 shares; Closing Stock Price: $19.49; Debt Value Outstanding 12/31/09: $618,100,000
- Weighted average cost of capital (WACC)—Using the target debt to value ratio of 6.99%, the WACC is approximately 10.14%.

**Valuation Assignment:** Your task is to estimate the intrinsic value of Toy Co. Inc.'s equity (on a per share basis) on 12/31/09 using the enterprise DCF model; this will assist you with determining what per share offer to make to Toy Co. Inc.'s shareholders. Treat all of the results/forecasts for the fiscal year ended 2010–2014 as projections. Your research on various historical merger and acquisition transactions suggests that comparable toy companies have been acquired at Enterprise Value/EBITDA multiples of 10.5×–11.5×. This is your assumption for a terminal-value exit multiple at the end of the forecast period, 2014. Exhibit 1 includes the target's planning period cash flow estimates, and Exhibit 2 provides market and other data for calculation of a weighted average cost of capital (WACC) for a discount rate.